

**PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA MINISTRY
OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH**



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**AI vs. Human Translation: A Case Study on Student Perceptions in
the English Department at the University of Oran 2**

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Dedication:

To the Almighty Allah source of infinite mercy, whose divine presence, immeasurable power, and boundless blessings have guided us to this moment and enlightened our minds with knowledge.

To our beloved parents, families, friends, and teachers for their steadfast support, enduring patience, and unconditional love that have sustained us throughout this journey.

To our younger selves whose perseverance in the face of adversity and commitment to the pursuit of education never wavered, even in the most uncertain of times.

To our future selves shaped by today's choices and nourished by the learning we embrace now, may you continue to grow with wisdom and purpose.

To all who live in this complex era where truth and illusion often blur, where the rapid advance of technology challenges the very essence of human identity. May we not lose ourselves in the noise but instead find clarity and meaning.

And finally, to the people of Gaza to the men, women, and children; to the martyrs whose lives were taken, and the brave who continue to resist; to the youth who got robbed of their right to learn. In your honor, we carry the sacred flame of knowledge a symbol of hope, remembrance, and unwavering solidarity with your angelic souls.

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Abstract

This dissertation investigates whether AI-powered translation tools can replace human translators, focusing on student perceptions at the English Department of the University of Oran 2. The first chapter presents a theoretical overview of both AI and human translation, exploring their definitions, historical development, methods, and key challenges particularly AI's limitations in context, nuance, and cultural meaning. The second chapter outlines the methodology, which involved a mixed-methods case study using a questionnaire completed by 32 Master's students. The third chapter analyzes the findings, showing that while AI is appreciated for its speed and convenience, students consistently prefer human translation for its accuracy, reliability, and emotional depth. The results suggest that AI should be seen as a support tool rather than a replacement, and that the future of translation lies in a collaborative model that blends human expertise with technological efficiency.

Keywords: AI translation, human translation, student perceptions, neural machine translation, translation quality, case study, University of Oran 2

ملخص

تهدف هذه الدراسة إلى معرفة ما إذا كانت أدوات الترجمة بالذكاء الاصطناعي قادرة على استبدال المترجمين البشر، وذلك بالاعتماد على تصورات طلبة قسم الإنجليزية بجامعة وهران 2. يتناول الفصل الأول عرضًا نظريًا حول الترجمة البشرية وترجمة الذكاء الاصطناعي، من حيث التعريف والنشأة والتقنيات، بالإضافة إلى أبرز التحديات التي تواجه الذكاء الاصطناعي مثل فهم السياق والدقة الثقافية والبعد العاطفي. يعرض الفصل الثاني منهجية البحث، التي اعتمدت على دراسة حالة باستخدام استبيان شمل 32 طالب ماستر. أما الفصل الثالث، فقد تناول تحليل النتائج، والتي أظهرت أن الذكاء الاصطناعي يحظى بتقدير من حيث السرعة، لكنه لا يزال أقل موثوقية من الترجمة البشرية، خاصة في النصوص الحساسة والدقيقة. توصلت الدراسة إلى أن الترجمة بالذكاء الاصطناعي لا يمكن أن تحل محل الترجمة البشرية، بل يمكن أن تدعمها، وأن مستقبل الترجمة قائم على التكامل بين الإنسان والآلة.

الكلمات المفتاحية: الترجمة بالذكاء الاصطناعي، الترجمة البشرية، تصورات الطلبة، الترجمة العصبية، جودة

الترجمة، دراسة حالة، جامعة وهران 2

Résumé

Ce mémoire examine si les outils de traduction automatisée peuvent remplacer les traducteurs humains, en se basant sur les perceptions des étudiants du département d'anglais de l'Université d'Oran 2. Le premier chapitre propose une vue théorique de la traduction humaine et de la traduction par IA, en abordant leurs définitions, leur évolution, leurs méthodes, ainsi que les limites de l'IA, notamment en matière de contexte, de nuance et de sens culturel. Le deuxième chapitre expose la méthodologie adoptée, fondée sur une étude de cas utilisant un questionnaire rempli par 32 étudiants en Master. Le troisième chapitre analyse les résultats, montrant que si l'IA est appréciée pour sa rapidité et sa praticité, les étudiants préfèrent la traduction humaine pour sa précision, sa fiabilité et sa profondeur émotionnelle. L'étude conclut que l'IA doit être vue comme un outil de soutien, et que l'avenir de la traduction repose sur une collaboration entre technologie et expertise humaine.

Mots-clés : traduction automatique, traduction humaine, perceptions étudiantes, traduction neuronale, qualité de la traduction, étude de cas, Université d'Oran 2

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List of Abbreviations:

(AI): Artificial Intelligence

(CAT): Computer-aided Translation

(NLP): Natural Language Processing

(NMT): Neural Machine Translation

(MT): Machine Translation

(SMT): Statistical Machine Translation

(RNNs): Recurrent Neural Networks

(ALPAC): the Automatic Language Processing Advisory Committee

(USAF): US Air Force

(EBT): Example-Based Translation

(TM): Translation Memory

(HT): Human Translation

(SL): The Source Language

(TL): The Target Language

General Introduction

General Introduction

Humans are social beings, and in order to communicate with others, they need means of communication that are sometimes incomprehensible, hence their incessant and rigorous search to find a way to understand and be understood. Among these tools, we have translation, which emerged as an activity as fascinating as it is complex in human thought, aiming to create connections between different communications in two or more languages, as well as their cultures. Beyond its complexity, translation stems from the fact that it aims to reconcile multiple languages and cultures. Therefore, translation is not exclusively the transfer from one language to another, but also the bridging of two or more cultures. This is how translation theories and approaches have emerged, aiming to produce translations as effective as possible, making translation a challenging task due to the differences and complexities that exist between languages.

In today's rapidly evolving technological landscape, the field of translation has undergone significant transformations. With the rise of artificial intelligence (AI), machine translation has gained widespread attention as an alternative to human translation. While AI-powered translation tools such as Google Translate and DeepL have demonstrated remarkable advancements in speed and accessibility, questions regarding their accuracy, contextual understanding, and overall reliability remain.

This thesis aims to explore the ongoing debate between AI translation and human translation, analyzing their strengths, weaknesses, and impact on the field of linguistics and translation studies. By conducting a case study at the Translation Department of the University of Oran 2, this research seeks to evaluate the practical implications of both translation methods in an academic setting.

1. Background and Context:

Translation has always played a crucial role in communication, bridging linguistic and cultural gaps across societies. Traditional human translation has been the cornerstone of this process, relying on linguistic expertise, cultural awareness, and contextual comprehension. However, with the advent of AI, translation has seen an unprecedented shift towards automation. AI translation is powered by machine learning algorithms and neural networks, enabling it to process vast amounts of text at remarkable speeds. Despite its efficiency, concerns persist regarding its ability to grasp linguistic nuances, idiomatic expressions, and cultural connotations. The increasing reliance on AI translation has sparked discussions among scholars, educators, and professionals in the field of translation studies. While some argue that AI can serve as a valuable tool to aid translators, others fear that it may compromise the quality and integrity of translation. This study aims to contribute to this discourse by assessing AI translation in comparison to human translation, focusing on key factors such as accuracy, fluency, and contextual awareness.

2. Research Problem:

The primary research question guiding this study is: How does AI translation compare to human translation in terms of accuracy, efficiency, and reliability? The study seeks to answer the following sub-questions: What are the strengths and limitations of AI translation compared to human translation? How do Master's students in the English Department at the University of Oran 2 perceive AI translation tools? What are the implications of AI translation for the future of professional translation? By addressing these questions, this research aims to provide a comprehensive analysis of the effectiveness of AI translation and its potential role in the translation industry.

3. Objectives of the Study:

This study is conducted with the following objectives:

1. To compare AI translation and human translation in terms of accuracy, fluency, and contextual comprehension.

2. To assess the impact of AI translation on translation education and professional practice.
3. To evaluate Master's students' perceptions at the Department of English, University of Oran 2, regarding the reliability and usability of AI translation tools.
4. To explore the future implications of AI translation for the translation industry.

4. Methodology:

The research adopts a mixed-methods approach, combining quantitative and qualitative data collection techniques within a single instrument. The study involves a structured questionnaire, comprising both closed- and open-ended items, administered to Master's students in the English Department at the University of Oran 2.

5. Significance of the Study:

This research is significant for several reasons. First, it contributes to the growing body of literature on AI translation, providing empirical data on its strengths and limitations. Second, it offers practical insights for translation educators and students, helping them understand the role of AI in translation training. Lastly, it informs professional translators and industry stakeholders about the evolving landscape of translation and the potential integration of AI into their work.

6. Structure of the Thesis:

This thesis is divided into Three main chapters:

- **Chapter One:** Provides an overview of AI translation and human translation, discussing their principles, methodologies, and theoretical frameworks.
- **Chapter Two:** Presents the case study conducted at the University of Oran 2, outlining the research methodology, data collection tools, and participant profiles, with a focus on the use of questionnaires and mixed-method analysis.

- **Chapter Three:** Analyzes and interprets the findings gathered from the questionnaire, offering both quantitative and qualitative insights into participants' experiences and perceptions. The discussion explores key themes such as accuracy, cultural nuance, trust, and the future role of AI in translation.

Through this research, the study aims to offer a balanced perspective on the AI vs. human translation debate, highlighting both the potential and the limitations of integrating AI into the field of translation. The findings contribute to the ongoing discourse on the future of translation, providing valuable insights for academics, professionals, and students alike.

7. Methodology Summary :

To explore the research questions, this study adopted a mixed-methods approach that combined both quantitative and qualitative data. The main tool used for data collection was a structured questionnaire distributed to Master's students in the English Department at the University of Oran 2. The questionnaire was designed to gather both statistical information (such as usage patterns and satisfaction levels) and open-ended reflections regarding their perceptions of AI and human translation.

8. Expected Results / Anticipated Outcomes :

It was expected that while students might appreciate the speed and accessibility of AI translation tools, they would still express a preference for human translation, especially in situations requiring emotional nuance, contextual awareness, and cultural sensitivity. The study anticipated that participants would view AI as a helpful tool but not yet capable of fully replacing human translators.

**CHAPTER 01 :
LITERATURE
REVIEW**

01

Introduction:

Recent years have seen huge developments in artificial intelligence that have an impact on various aspects of life. This has specifically affected text machine translation, which has reduced or eliminated human contact the thing that led to the widespread availability of translation software models based on artificial intelligence (AI), such as Google Translate, Bing, Microsoft Translator, DeepL, Reverso, Systran Translate, and Amazon Translate. There are also several computer-aided translation (CAT) tools available, including Memoq, Trados, Smartcat, Lokalise, Smartling, Crowdin, TextUnited, and Memsource. Artificial intelligence has been used more recently to create programs like ChatGPT, ChatSonic, GPT-3 Playground, Chat GPT 4, and YouChat that more closely resemble human interactions by simulating conversational answers to researchers' questions.

This chapter aims to provide a comprehensive introduction to both AI translation and human translation by exploring their fundamental principles, methodologies, and theoretical foundations. It seeks to define and differentiate these two translation approaches, highlighting their strengths, limitations, and key areas of application. By examining the evolution of AI translation and the role of human translators, this chapter will establish a foundation for understanding how these methods coexist, complement each other, and continue to shape the field of translation studies.

1.1 AI Translation Definition:

Artificial Intelligence (AI) Translation refers to the use of deep learning, neural networks, and natural language processing (NLP) to automate the translation of text or speech from one language to another. Unlike traditional rule-based or statistical approaches, AI translation systems rely on large-scale neural machine translation (NMT) models, which learn linguistic patterns from vast datasets to produce more context-aware, fluent, and human-like translations. According to Bahdanau et al. (2015), AI translation, particularly in the context of NMT, is characterized by its ability to jointly learn to align and translate words using an encoder-decoder framework. This allows the model to capture context dependencies between words, improving accuracy and reducing literal translations. Further advancements in AI translation were made with the introduction of Transformer models (Vaswani et al., 2017), which utilize self-attention mechanisms to process entire sentences simultaneously rather than word by word. This innovation significantly enhanced fluency, coherence, and handling of long-range dependencies, making modern AI translation systems highly effective. As Koehn (2020) points out, AI translation represents a shift toward more sophisticated models that not only translate words but also grasp semantic meaning, idioms, and cultural nuances. These systems continuously improve through reinforcement learning and fine-tuning, adapting to new languages and domains more efficiently than previous approaches. AI, NLP and machine translation are closely related together which relationship is shown in **Figure 1.1**

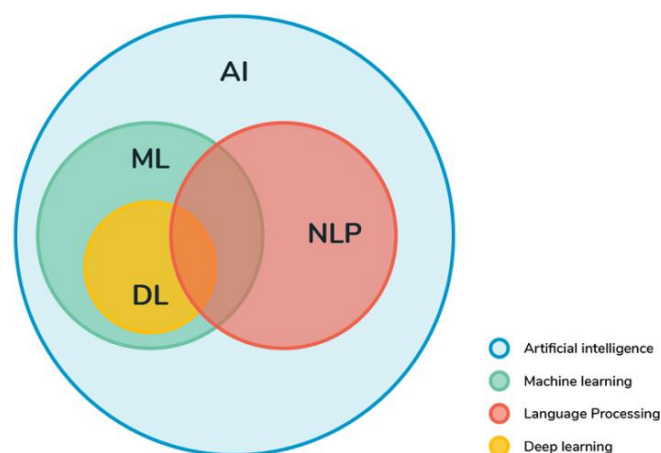


FIGURE 1.1: THE RELATIONSHIP BETWEEN AI, NLP AND MACHINE TRANSLATION

1.2 History of AI Translation:

Machine Translation (MT) has come a long way since it was first imagined in the mid-20th century. The idea was to use computers to automatically translate languages, but progress was slow and faced many challenges. Over time, new methods improved translation quality, moving from simple word replacement to complex AI-driven systems. Understanding the history of MT helps us see how technology has evolved and how it continues to shape communication today. The following sections highlight the key moments in this journey. **Figure 1.2**

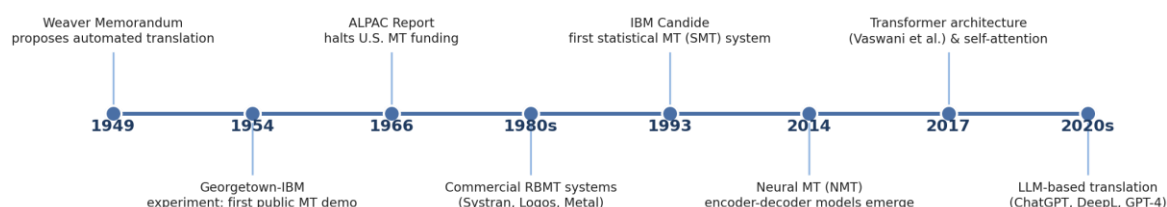


Figure 1.2: Timeline of Key Milestones in Machine Translation History (1949–2020s)

1.2.1 The Early Years of Machine Translation (1947-1954) :

Shortly after the emergence of electronic computers, researchers began exploring their potential for translating natural languages. The origins of this idea can be traced back to March 1947, when Warren Weaver from the Rockefeller Foundation corresponded with cyberneticist Norbert Wiener about the possibility of automated translation. Two years later, in July 1949, Weaver composed a memorandum outlining various ideas, drawing inspiration from advancements in codebreaking during World War II, Claude Shannon's work in information theory, and the notion that universal principles might govern all human languages. By the early 1950s, machine translation (MT) research had gained traction across several universities in the United States. In 1954, IBM and Georgetown University conducted the first public demonstration of machine translation. Although the system operated with a limited vocabulary and grammatical structure, its results were compelling enough to secure substantial funding in the U.S. and inspire the creation of similar projects worldwide.

1.2.2 The Era of Optimism (1954-1966):

During the early years of machine translation (MT), systems primarily relied on extensive bilingual dictionaries, where each word in the source language was matched with one or more equivalents in the target language. Additionally, basic rules were implemented to ensure proper word order in translations. However, researchers soon realized that relying solely on dictionary-based rules for syntactic structuring was impractical, as they became increasingly complex and inconsistent. This recognition led to a shift toward more structured approaches to syntactic analysis, inspired by contemporary linguistic theories, such as generative-transformational grammar, dependency grammar, and stratificational grammar. These frameworks were believed to hold the potential for significantly improving translation accuracy. Throughout the first decade, optimism remained strong, with many researchers anticipating major breakthroughs. However, as the field progressed, semantic challenges emerged, proving far more difficult to overcome than initially expected. While some operational systems were developed such as the Mark II system, created by IBM and Washington University for the USAF Foreign Technology Division, and the Georgetown University system, used by the US Atomic Energy Authority and Euratom in Italy, their output quality was underwhelming. Despite this, they were still valuable for users needing quick access to translated information. By 1964, frustration over the slow progress of MT research led the US government to establish the Automatic Language Processing Advisory Committee (ALPAC). In its 1966 report, ALPAC concluded that machine translation was slower, less accurate, and twice as costly as human translation, with no foreseeable improvements that would justify further investment. Instead, the committee recommended shifting focus to machine assisted translation tools, such as automatic dictionaries, and continuing fundamental research in computational linguistics rather than full-scale machine translation development.

1.2.3 The Impact of the ALPAC Report (1966-1980):

Despite facing widespread criticism for being biased and overly pessimistic, the ALPAC report led to a significant decline in machine translation (MT) research in the United States for more than a decade. Its influence also extended to the Soviet Union and Europe, where funding and interest in MT development were affected. However, research efforts continued in other regions, particularly in Canada, France, and Germany. A few years later, MT systems began to resurface in practical applications. In 1970, the US Air Force (USAF) 13 adopted the Systran system, and by 1976, the Commission of the European Communities implemented it to handle the growing volume of multilingual documents. That same year, another notable MT system, Meteo, was introduced in Canada by researchers at Montreal University to automate the translation of weather reports. During the 1960s, both the United States and the Soviet Union focused primarily on Russian English and English Russian translation, particularly for scientific and technical documents. The target audience for these systems was limited, and users were willing to accept unpolished, raw translations in exchange for faster access to information. However, by the mid-1970s, the demand for MT had shifted. New users, including multilingual administrative bodies and global commercial enterprises, required translation solutions that could handle varied languages and large volumes of documentation. This increasing demand for cost-effective, machine assisted translation systems was particularly evident in Europe, Canada, and Japan, where international trade and multilingual communication had exceeded the capacity of traditional human translation services. Consequently, the focus of MT development evolved to address the commercial and technical documentation needs of global business and administration.

1.2.4 The Development of Machine Translation in the 1980s:

The 1980s saw a significant expansion in machine translation (MT) systems, with a growing number of countries contributing to their development. During this period, several mainframe-based MT systems were introduced and remained in use for many years. Notable among these were Systran, which supported multiple language pairs, Logos (German English, English French), the Pan American Health Organization's internal system (Spanish English, English Spanish), the Metal system (German English), and several Japanese-developed systems for translating between English and Japanese. The advent of microcomputers and text-processing software created a demand for more affordable MT solutions, leading to the rise of various commercial systems. In North America and Europe, companies such as ALPS, Weidner, Linguistic Products, and Globalink entered the market, while in Japan, major technology firms including Sharp, NEC, Oki, Mitsubishi, and Sanyo developed their own MT software. Additionally, microcomputer-based MT systems emerged in China, Taiwan, Korea, Eastern Europe, and the Soviet Union, further expanding the field.

Throughout the decade, research in MT continued to evolve, focusing on more sophisticated translation techniques. The dominant approach involved 'indirect' translation, utilizing intermediary linguistic representations. Some systems adopted an interlingual approach, incorporating semantic, morphological, and syntactic analysis, while others integrated non-linguistic knowledge bases to enhance translation accuracy. Several major projects emerged during this time, including GETA-Ariane (Grenoble), SUSY (Saarbrücken), Mu (Kyoto), DLT (Utrecht), and Rosetta (Eindhoven). Additionally, knowledge-based MT research was conducted at Carnegie Mellon University (Pittsburgh). Two major international multilingual initiatives also took shape: Eurotra, backed by the European Communities, and the Japanese CICC project, which involved collaboration with China, Indonesia, and Thailand.

1.2.5 Advancements in Machine Translation in the 1990s:

The early 1990s marked a significant turning point in machine translation (MT) due to several groundbreaking developments. First, IBM researchers introduced *Candide*, a system that relied entirely on statistical methods rather than traditional linguistic rules. Around the same time, Japanese research groups pioneered the use of example-based translation (EBT), which involved leveraging translation corpora instead of syntactic or semantic rules to analyze texts or select lexical equivalents. Both approaches were distinct from earlier rule-based methods, as they relied on large scale text databases rather than manually crafted linguistic rules. Despite the rise of these new approaches, traditional rule-based MT projects continued. Notable examples included the Catalyst project at Carnegie Mellon University, the University of Maryland's research, and the ARPA-funded Pangloss project, which involved collaboration between three U.S. universities. Another major innovation was the introduction of translation memory (TM) systems, such as Trados, which provided translators with quick access to previously translated content, significantly improving efficiency. A fourth significant advancement was the initiation of research in speech translation, which aimed to integrate speech recognition, synthesis, and machine translation. These systems often combined rule-based and corpus-based approaches. Some of the most notable projects in this field included ATR (Nara, Japan), the JANUS project (a collaboration between ATR, Carnegie Mellon University, and the University of Karlsruhe), and Verbmobil, a German government-funded initiative. Finally, the early 1990s also saw a shift toward practical applications of MT. Researchers focused on developing translator workstations for professionals, refining controlled language and domain-specific translation systems, and incorporating MT components into multilingual information systems. The trends from the early 1990s continued to shape the field of machine translation (MT) throughout the late 1990s. One of the most notable developments was the rapid expansion of MT tools and translation aids, such as translator workstations and translation memory (TM) systems, particularly among large corporations. A significant area of growth was software localization, which involved adapting and

translating software documentation to suit different linguistic and cultural markets. During this period, there was also a surge in demand for MT software designed for personal computers, primarily targeting non-professional users. This demand was met not only by newly developed systems but also by downsized and enhanced versions of existing mainframe-based MT software. While some applications required moderately high translation quality, particularly for published content, another sector experienced even faster growth: automatic translation for the Internet. This included translation for emails, web pages, and online communication, where speed and real-time response were prioritized over translation accuracy. A crucial milestone for the future of MT was the introduction of online translation services, starting with Babelfish and later followed by Google Translate. These advancements transformed MT into a widely accessible, mass-market technology, making automated translation tools available to the public on a large scale.

1.2.6 Machine Translation Since 2000:

Since the turn of the century, Statistical Machine Translation (SMT) has emerged as the leading approach in MT research, with rule-based methods becoming increasingly rare. Several factors have contributed to this shift:

1. The availability of extensive bilingual and monolingual corpora.
2. The open-source development of essential SMT tools like Moses and GIZA, which facilitate tasks such as alignment, filtering, and reordering.
3. The establishment of widely recognized evaluation metrics like BLEU and its successors, which allow for systematic assessment of MT systems. Unlike rule-based translation, SMT does not require researchers to have deep linguistic expertise in the languages being translated, nor does it necessitate the complex and large-scale manual acquisition of grammatical rules and lexical data. However, while SMT remains dominant, rule-based methods still play a crucial role in addressing specific challenges. These include sentence reordering for languages with significantly different structures (e.g., English and Japanese), handling morphologically rich

Chapter 1: Literature Review

languages (such as Russian and Finnish), managing transliterated names (especially in Chinese), and improving discourse coherence, such as pronoun resolution. As a result, many researchers now combine SMT with rule-based techniques, leading to hybrid MT models. The use of MT has significantly expanded, reaching new domains such as patent translation, TV subtitles, website localization, restaurant menus, social media, humanitarian efforts, corporate communications, and public lectures. Large multinational corporations and translation services increasingly integrate MT with pre-processing techniques (e.g., controlled language and terminology checks) and 20 post-editing strategies, including crowdsourced corrections and statistical refinements. Unlike in the past, professional translators have become more accepting of MT, utilizing it alongside translation memories to assist in draft translations. For the public, access to MT is now predominantly through free online services like Google Translate, rather than standalone PC software. Although translation quality is perceived to be improving, both in general online MT systems and specialized research projects, evaluation remains a critical area of focus. Notably, human assessments of translation quality often differ from statistical evaluation metrics, highlighting the need for more effective ways to measure MT accuracy and performance.

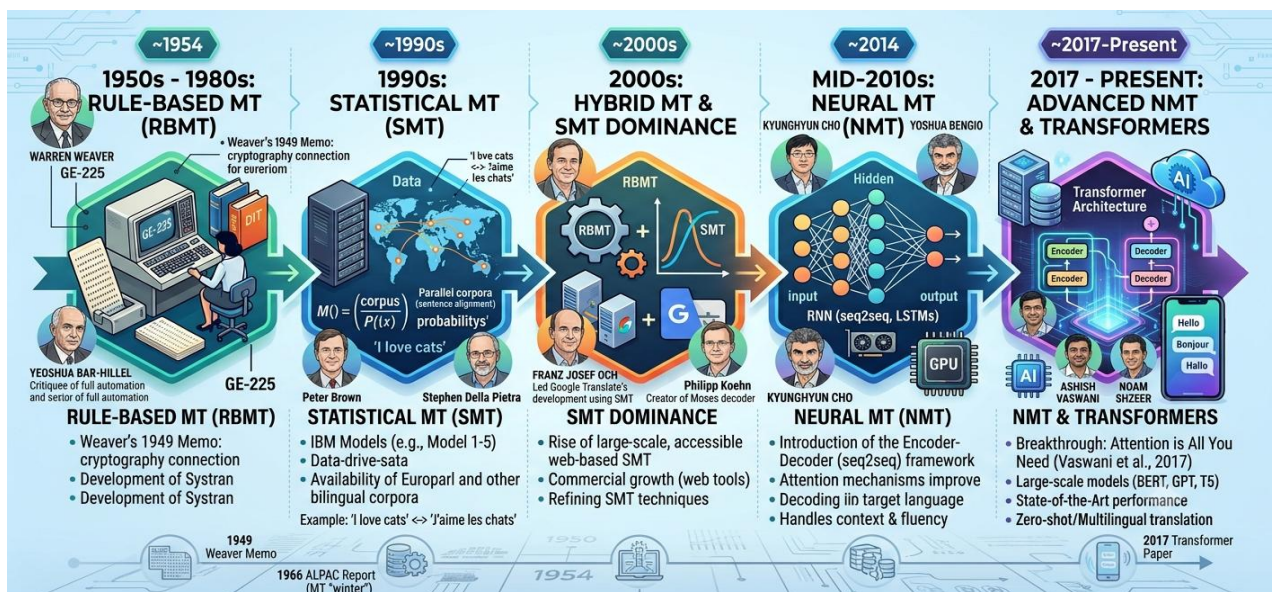


FIGURE 1.3 : THE EVOLUTION OF ARTIFICIAL INTELLIGENCE AND MACHINE TRANSLATION

1.3 Technological foundations:

This chapter explores the different methods and key trends in machine translation from its early days until now. Before going into detail, it is important to understand the main challenges and developments in the field.

1.3.1 Rule-Based Machine Translation :

From Direct to Interlingual Methods Machine translation has evolved through different approaches, each using unique techniques to process and translate text. Some systems translate text directly from one language to another without any intermediate steps, while others take a more structured approach by analyzing the meaning before generating the translation. In some cases, this intermediate step is designed to be language independent, allowing the same source text to be translated into multiple target languages. To make things simpler, we can categorize these approaches into three main types, commonly found in textbooks on the subject:

a. Direct Translation Systems

These systems work in the most straightforward way translating words from one language to another without any deeper analysis. They rely heavily on dictionaries that provide word-for-word translations, followed by a set of basic rules to rearrange words so they better match the grammar of the target language. However, because they do not analyze sentence structure, the translations can often sound unnatural.

b. Transfer-Based Systems

Unlike direct translation, transfer-based systems take the sentence structure into account. They first analyze the syntax of the source language, then apply rules to transform it into a more natural sentence in the target language. This approach avoids the rigid word-for-word limitation of direct systems, resulting in translations that are more fluent and idiomatic provided the system accurately captures the grammar and structure of both languages.

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c. *Interlingual Systems*

The most advanced and ambitious approach involves creating an abstract representation of the text before translating it. This "interlingua" is designed to be language-neutral, capturing the meaning in a way that allows the content to be translated into multiple languages from the same source. Researchers have debated how precise and detailed an interlingua should be to accurately represent a sentence. Instead of developing an entirely new artificial language for this purpose (which is extremely complex), some systems use an existing language often English as a "pivot." In this case, the system first translates the text into the pivot language and then from there into the final target language. However, this method is not truly language-independent and has its own challenges. These three approaches exist on a spectrum, from simple word-for-word translation to highly sophisticated models that aim to break free from language-specific structures. Each method has its strengths and limitations, shaping the evolution of machine translation over time. They are often illustrated using a well-known diagram called the "Vauquois Triangle," named after a prominent French researcher in machine translation from the 1960s. This visual representation effectively summarizes the various approaches and how they relate to one another.

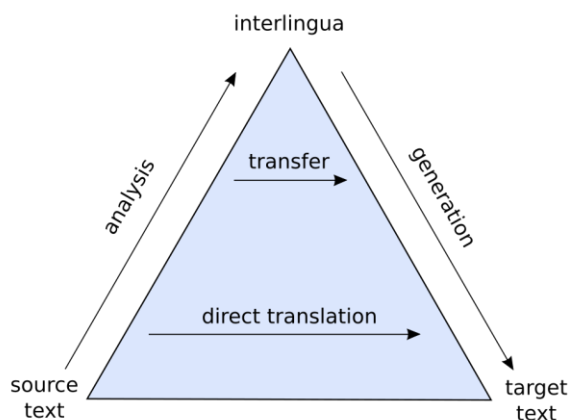


Figure 1.4: The Vauquois Triangle – Direct, Transfer-Based, and Interlingual Translation Approaches

At the base of the Vauquois Triangle lies direct transfer, which represents the most basic form of translation word-for-word conversion. This approach requires no deep analysis of the source text, often relying on a simple bilingual dictionary. However, because languages differ significantly in structure and expression, this method is ineffective. While it may offer a rough understanding of a text, its usefulness is mostly limited to cases where the languages are closely related, such as those within the same language family or those with similar syntax. From the very beginning, researchers recognized the limitations of direct translation and sought to develop more refined methods that take linguistic structure into account. This led to the introduction of transfer rules in the 1950s. Instead of translating individual words in isolation, these rules focus on translating linguistic units, such as phrases and idiomatic expressions. Since sentence structures vary significantly between languages, direct sentence-level translation is impractical. Instead, sentences can be broken into smaller chunks that follow predictable patterns. For instance, in French, adjectives often come after the noun, while in English, they typically appear before it. Transfer rules help accommodate such differences. More complex transformations are also necessary for example, converting the French phrase "*je veux qu'il vienne*" into English as "*I want him to come.*" A direct word-for-word translation ("*I want that he comes*") sounds unnatural in English, just as "*je veux lui de venir*" is incorrect in French.

Beyond syntax, transfer rules can also be applied at the semantic level to help determine the correct meaning of a word based on context. Consider the word "bank." Depending on the sentence, it could refer to a financial institution or the side of a river. Manually defining all contexts for a word is extremely difficult, making this one of the biggest challenges in early machine translation. More recently, however, statistical methods have significantly improved accuracy by leveraging large datasets an area where computers excel far beyond human capabilities, especially when it comes to creating explicit, formal models for such problems.

1.3.2 The Interlingual Approach:

A More Ambitious Solution Another major approach, interlingual translation, aims to go beyond the limitations of transfer-based systems. Transfer rules are always specific to a particular language pair, meaning a separate set of rules must be developed for every new combination (e.g., English to French, English to Spanish, etc.). Interlingual systems attempt to solve this issue by introducing an abstract, language-independent representation of meaning. Instead of directly translating between languages, a system first converts the source text into this intermediate representation, which can then be used to generate translations in multiple target languages. For this approach to work, a generation module is required a system capable of taking the abstract interlingual format and transforming it into grammatically correct sentences in different languages. This method is highly ambitious because it demands both a deep understanding of the input text and the ability to generate accurate, natural-sounding sentences in multiple languages. However, despite decades of research, large-scale deployment of interlingual systems has remained elusive. One of the biggest hurdles is the concept of understanding what does it truly mean for a machine to "comprehend" a text? How much information must be captured to ensure an accurate translation? And to what extent can the comprehension process be formalized? Given the complexities involved, achieving a fully functional interlingual system remains beyond the current capabilities of artificial intelligence. Understanding a text involves not only explicit meaning but also inferred context and deeper layers of interpretation an extraordinarily difficult challenge that continues to push the boundaries of modern machine translation research.

1.3.3 The Rise of Statistical Machine Translation Since the early 1990s:

The field of machine translation has undergone a major transformation, challenging traditional classification methods. This shift was driven by the growing availability of massive amounts of text especially with the rise of the internet and the increasing power of computers. These advancements paved the way for statistical machine translation (SMT), which took a completely different approach from earlier rule-based systems. Unlike traditional methods that relied on extensive bilingual dictionaries and manually crafted translation rules, SMT systems learn translations from large-scale bilingual data. Initially, these systems operated similarly to direct translation models, identifying word equivalents by analyzing vast amounts of parallel text. Many of these texts originally came from international organizations, but modern systems now gather much of their data from the web. Over time, statistical translation methods have evolved to handle more than just single-word translations. Instead of treating words in isolation, modern SMT models recognize multi-word phrases, idiomatic expressions, and set word patterns that must be translated as a unit. The latest versions even attempt to translate at the sentence level, making them more advanced than their earlier word-based counterparts. However, these systems generate internal representations of language that are not easily interpretable by humans. This raises key questions: Do these models truly understand meaning? Can their approach to language be compared to human cognition? One of the main reasons for the success of statistical systems is their ability to identify patterns and recurring structures in language purely through probability calculations. Since meaning is derived from how words are used in context, SMT can detect common translation patterns and regularities without needing explicit rules. However, these systems have long struggled with assembling coherent sentences, as they often rely on overlapping fragments that must be pieced together. While statistical approaches have significantly improved machine translation, they still face limitations, particularly in producing fluid and contextually accurate sentences. The next step in this evolution involves refining how these fragments are processed and integrated, bringing translation technology closer to human-like understanding.

1.3.4 Deep learning:

Deep learning has transformed machine translation by leveraging neural networks to process language in a structured way, like how the human brain forms complex concepts. Unlike traditional rule-based methods, deep learning automatically identifies patterns and relationships in language, making it more adaptable to the complexities of human communication. Its potential was widely recognized in 2016 when Google DeepMind's AlphaGo, powered by deep learning, defeated the world champion in Go. While human language presents unique challenges such as ambiguity, context dependence, and cross language differences deep learning offers a promising solution by efficiently handling these complexities, pushing machine translation to new levels of accuracy and fluency. Deep learning first gained success in image recognition, where it autonomously extracts relevant features from vast datasets instead of relying on predefined characteristics. It follows a hierarchical learning process, starting with basic elements (pixels, words) and gradually forming more complex structures (images, sentences). In machine translation, deep learning reduces manual intervention by allowing the system to infer optimal representations from data. Unlike earlier statistical models that required manual parameter adjustments, deep learning models neural machine translation (NMT) use an encoder-decoder architecture to process entire sentences rather than breaking them into fragments. This method captures both contextual and syntactic relationships, improving translation fluency and accuracy. A key feature of deep learning is word embeddings, where words with similar meanings are grouped based on their contextual usage rather than rigid definitions. This continuous representation makes translations more adaptable and precise, enabling the system to handle paraphrases, synonyms, and ambiguous terms more effectively. Despite its advancements, deep learning remains computationally demanding, requiring specialized hardware and extensive training time. Additionally, neural networks' architecture lacks a strong theoretical foundation, with many design choices based on empirical results rather than established linguistic principles. While the approach has significantly improved image recognition, speech processing, and natural language understanding, challenges remain especially in grasping complex sentence

structures, which can still result in translation errors. Nonetheless, deep learning continues to push the boundaries of machine translation and AI-driven language processing. Initially, deep learning-based machine translation systems excelled with simple sentences but struggled with complex ones. This was due to challenges such as the high complexity of training neural networks, difficulty handling unknown words, and the omission of certain word groups in translations. These limitations hindered their commercial deployment for a time. However, recent advancements have improved efficiency. Optimization techniques have simplified training, and attention mechanisms now enhance the connection between the encoder and decoder, allowing the model to focus dynamically on the most relevant words in a sentence. This is particularly beneficial for long or structurally different languages, such as English and Japanese. The issue of unknown words remains significant, as deep learning models can only translate words present in their training data. Unlike modular statistical systems that addressed this issue more easily, neural models require alternative strategies. Some approaches include copying proper nouns or numbers from the source text or using transliterations for different writing systems. Another method involves breaking unknown words into smaller units for context clues, but this solution is not fully reliable. Additionally, deep learning models sometimes fail to translate entire sentences accurately, omitting words or phrases due to the holistic nature of their processing. To counteract this, Google introduced a length penalty mechanism that prioritizes longer, more complete translations. However, since deep learning models operate on numerical representations, understanding their internal decision-making remains a challenge. Research efforts now aim to make these processes more interpretable. Deep learning has significantly advanced machine translation, first improving short sentences in related languages and now extending to longer sentences and more diverse language pairs. The rapid adoption of this approach by major industry players (Google, Bing, Facebook, Systran, etc.) suggests its robustness, with deployment occurring much faster than the shift from rule based to statistical methods. Despite these advancements, the approach is still evolving, with ongoing improvements expected, particularly in handling unknown words. Some researchers

advocate for hybrid models, integrating neural networks with traditional statistical methods for better precision. However, this contrasts with the deep learning philosophy of processing sentences holistically. In conclusion, while deep learning has revolutionized translation, it has yet to achieve human-level accuracy. Unlike chess or go, where AI has surpassed human champions, natural language processing remains a complex challenge.

1.4 Challenges in Ai translation:

Despite significant advances in AI-based translation, there are still several challenges and limitations. The topic of AI translation is quickly evolving and has the potential to completely change how we interact between languages. Before AI translation can be utilized to translate all forms of text accurately and reliably, there are still several issues that need to be resolved.

1.4.1 Ambiguity:

It is the core challenge in natural language processing. It exists at all levels, including word meaning, morphology, syntax, and textual relationships. While humans navigate ambiguity using context and background knowledge, misunderstandings still occur. Sometimes, people are intentionally unclear, and in those cases, translations should keep the ambiguity instead of choosing a single meaning. A common issue in translation is that some words have multiple meanings. For example:

- She put the bat in her sports bag after practice.
- A bat flew out of the cave at sunset.

In the first sentence, "bat" refers to sports equipment, while in the second, it means a flying animal. Similarly, the word "bright" could mean intelligent in one context or describe strong light in another. How do people tell the difference? They rely on surrounding words and the overall meaning of the sentence. For instance, the phrase "after practice" suggests the sports-related meaning of "bat." Computers must also analyze context to correctly interpret and translate words.

1.4.2 Challenges in Translating Phrases:

A major difficulty in translation is that meaning is not always built word by word. This makes it hard to break translation into smaller, independent tasks. One of the clearest examples is idioms. For instance, the English phrase "kick the bucket" does not actually refer to kicking a bucket it means someone has passed away. A good French equivalent would be "casser sa pipe," which means "break one's pipe" but conveys the same meaning.

While some idioms can be traced back to historical events or metaphors, most people simply learn them without analyzing their origins. Since idioms do not translate directly, machine translation systems must recognize and replace them with culturally appropriate equivalents instead of translating word for word.

1.4.3 Challenges in Syntactic Translation:

A common issue in translation is syntactic ambiguity, particularly with prepositional phrases. Consider the difference between "He painted the house with a brush" and "He painted the house with his brother." In the first sentence, "with a brush" describes how he painted, while in the second, "with his brother" describes who helped him. However, this type of ambiguity does not always pose a problem in translation, as some languages allow similar ambiguous structures. More significant challenges arise from differences in sentence structure between languages. Some languages, like English, rely on word order to indicate relationships between words, following a subject-verb-object pattern. Others, like Latin or Russian, use word endings (morphology) to clarify roles, allowing for more flexible word order. These structural differences must be carefully considered to ensure accurate translation.

1.4.4 Challenges in Semantic Translation:

Translation becomes especially difficult when meaning is conveyed differently across languages or when understanding requires inference beyond literal words. One major issue is pronoun reference (pronominal anaphora) when pronouns refer to previously mentioned words. For example, in the sentence: "I read the book, and it was fascinating." Here, "it" clearly refers to "book." However, in languages like Spanish or Italian, nouns have gendered forms. In Spanish, "book" translates to "libro," which is masculine, so "it" must be translated as "él" rather than a neutral pronoun. Similarly, other nouns may have different gender assignments in different languages "bridge" is masculine in Spanish (el puente) but feminine in German (die Brücke). This process requires multiple layers of understanding: identifying the connection between "it" and "book," selecting the correct translation, recognizing the noun's gender, and ensuring the pronoun matches. This highlights the challenge of co-reference resolution, where translation systems must correctly track and interpret references across sentences.

1.4.5 Corpora and Machine Translation:

Types of Text : Machine translation (MT) systems are trained using translated texts. However, there is no one-size-fits-all system. Different text types require different approaches. For example, an MT system that performs well with medical research papers might struggle with casual conversations in online chatrooms. Words and phrases change meaning depending on the subject, and translations must match the specific style of each domain. MT is used for both written and spoken language. Spoken content must be converted into text before translation, either through manual transcription (which may involve removing unnecessary words) or through speech recognition software. However, spoken language is often unstructured, full of incomplete sentences, pauses, and references to shared knowledge. This is also true for informal written communication, such as emails, text messages, and social media posts. Another important factor is the topic of the text. Many available translated texts come from global organizations like the United Nations, covering political and economic subjects. However, these may not be useful for training MT systems in specialized areas like scientific research. Historically, MT was developed for specific

domains, such as weather reports or travel information. Many valuable sources of translated content, such as books and product manuals, are not publicly available, limiting access for researchers. However, companies with extensive internal translations, such as product documentation from previous years, can build their own MT systems tailored to their needs.

Acquiring Parallel: Texts A parallel corpus is a collection of texts paired with their translations in another language. Many large-scale parallel corpora, containing millions of words, are now accessible through sources like the Linguistic Data Consortium at the University of Pennsylvania. Another way to collect parallel data is by searching the web, as many websites and news organizations provide content in multiple languages. However, gathering parallel texts from the web can be challenging. First, extracting text from websites requires technical tools to download and process HTML content. Next, the challenge is document alignment identifying which texts in different languages correspond to each other. Sometimes, website structures or URLs provide clues, but often, translations are not exact. Many news websites, for example, do not directly translate articles but instead rewrite them to suit different audiences. In such cases, only portions of the content may be direct translations, making alignment more complex. Apart from direct translations, researchers also study comparable corpora, which contain texts in different languages covering the same topics but are not exact translations. For instance, news archives from English and French newspapers often discuss the same events but with different wording. By analyzing these comparable texts, researchers can identify translation patterns based on how words appear in similar contexts. However, direct translations remain the strongest resource for training MT systems.

Sentence Alignment: Even when a parallel corpus is available, further processing is needed to make it usable for statistical machine translation. A key step is sentence alignment, where corresponding sentences in different languages are matched. Translations are rarely word-for-word or even sentence for-sentence. Some languages break long sentences into multiple smaller ones, while others merge short sentences into longer ones. Some languages, like Thai, do not even have clear sentence

boundaries in writing. Since sentence alignment is crucial for training MT systems, researchers have been working on it for a long time, developing algorithms to accurately match corresponding sentences. This step ensures that MT models can effectively learn translation patterns from parallel texts.

1.5 Human translation Definition:

(HT), by definition, is when a human translator- not a machine- translates the text. It is the oldest form of translation and relies on pure human intelligence to transform one way of saying things into another. And while human translation comes in many different flavors, it remains the most widely used translation method to this day. Furthermore, the word translation can mean different things. It can refer to the field of study related to translating, the final translated text or the process of converting a text from one language to another. When translating between two written languages, a translator takes the original text (source text) in one language (source language) and converts it into a new text (target text) in another language (target language). This type of translation is called interlingual translation one of three types identified by linguist Roman Jakobson in his influential paper *On Linguistic Aspects of Translation* (1959/2004). Jakobson classified translation into three categories:

1. Intralingual translation (Rewording) Rewriting a message in the same language using different words. For example, explaining a difficult concept in simpler terms.

2. Interlingual translation (Translation Proper) Converting a text from one language to another. This is the most common form of translation.

3. Intersemiotic translation (Transmutation) Changing a message from a verbal form into a non-verbal system. For example, turning a novel into a movie or a poem into a painting.

1.6 Traditional translating methods:

Since there is no best way to translate, translators gradually came up with their own methods that improve both the accuracy and speed of their translations. Although no single set of guidelines applies to everyone, there are certain techniques and tools for translation that can help practically any translator produce work with higher accuracy. In his book "A Textbook of Translation" Peter Newmark and other academics discuss a number of translation techniques, including word-for-word translation, literal translation, faithful translation, communicative translation, semantic translation and adaptation. These will be described below, with examples to show each type.

1.6.1 Word-for-word translation:

This type of translation keeps the source language (SL) word order; words are translated out of context according to their most common meaning. Such kind of translation can be used as a preliminary translation step, but it is not applied in real translation tasks. This method or type of translation takes the meaning of each word in isolation regardless of differences between both languages in grammar, word order, context, and special usage. Moreover, this translation focuses on the source language, and the target should follow it step by step. Hence, it seems a very easy way to translate, and it is common between students. However, this method is very risky because it does not consider the target language and relies on the source language only.

1.6.2 Literal translation:

Literal translation refers to a method where the source text (ST) is translated word-for-word without considering its context or possible shifts in meaning, often leading to a loss in the communicative value of the translation. This approach can distort the original message, especially when the source text's meaning doesn't align with the target language's (TL) conventions. However, literal translation can work when the languages share identical expressions, grammatical structures, or straightforward terms. Sometimes, literal translations fail when trying to preserve a deeper meaning. Literal translation often arises from choosing incorrect translation units, where the words or expressions don't align between languages. This method of translation resembles the first one (word-for-word translation) in two main aspects:

- (a) It considers the source language word order.
- (b) It emphasizes on having the same kind and number of words.

However, it differentiates from it in two points:

- (a) It does not neglect context.
- (b) It finds metaphorical equivalents in the target language for metaphorical words in the source language.

So, this method of literal translation is much more acceptable than the first method, which insists on having a word-for-word translation.

1.6.3 Faithful translation:

This method maintains a balance between the literal meaning of the source language (SL) word and the target language (TL) syntactic structures. A faithful translation attempts to reproduce the precise contextual meaning of the original within the constraints of the target language's grammatical structures. It 'transfers' cultural words and preserves the degree of grammatical and lexical abnormality in the translation. It attempts to be completely faithful to the intentions and the text realization of the source language writer. It sounds more reasonable as it takes the context into consideration, aiming at producing a more precise meaning of the SL texts.

1.6.4 Communicative translation:

This type of translation attempts to keep the exact contextual meaning of the original text in such a way that both content and language are readily acceptable and comprehensible to the reader. It is particularly suitable when translating conventional formulae or proverbs, and it involves some levels of cultural approximation. Communicative translation aspires to create the same effect created by the SL text on the TL reader. Though it is not as accurate as semantic translation, which sticks to the original text, it communicates the meaning at the expense of accuracy. However, it is preferred by many translators because it resorts to concepts that are more familiar to the TL reader on cultural and social levels.

1.6.5 Semantic translation:

Semantic translation attempts to retain, as closely as the semantic and syntactic structures of the second language allow the exact contextual meaning of the original. Semantic translation aims at replicating the original texts' forms within the target language, reproducing the original context, and retaining the characters of the SL culture in the translation. Although not necessarily a literal translation, it follows the source text more closely.

In contrast, communicative translation centers on the specific language and culture and focuses on the TL readers. It is focused on the target text and aims to ensure that the reader will understand the message of the text.

Semantic translation, on the other hand, differs from 'faithful translation' only in as far as it must take more account of the aesthetic value of the source language text, compromising on the 'meaning' where appropriate so that no assonance, wordplay or repetition remains in the finished version.

1.6.6 Adaptation:

Adaptation means the modification of the idea in the source language (SL) so as to find an acceptable one in the target language (TL). It is necessary when something specific to one language culture is expressed in a totally different way that is familiar or appropriate to another language culture. It is a shift in cultural environment.

In other words, adaptation is a kind of rewriting of the ST to make it conform to the rules of the language and especially the culture of the TL community. It is considered the freest form of translation and is used mainly for plays (comedies) and poetry. In addition, the translation of metaphors, proverbs, etc., also frequently involves this technique.

In this way, the procedure of adaptation aims at achieving an 'equivalence' of situations by considering a similar context in the TL whenever a cultural gap between ST and TT defies comprehension. Thus, adaptation can be considered an instance of equivalence at the level of context.

1.6.7 Free translation:

This method preserves the meaning of the original but uses natural forms of the TL, including normal word order and syntax, so that the translation can be naturally understood. It preserves the content at the expense of the form, and it provides a longer paraphrase of the original. It is a form of idiomatic translation that favors colloquialisms and idioms that do not exist in the SL.

In other words, this type of translation focuses on translating freely because the translator is not limited to the text or context or the denotative meaning of a word or a phrase but goes beyond words and phrases and out of texts and outside contexts. So, the only restriction depends on the translator's comprehension.

1.7 The role of human translators:

It is arguable that translation has always been examined as a human activity. Translation professionals first considered the effects of translation on their communities or on their own translations. So why did translation academics feel compelled to begin writing about translation as a human activity in the 1990s? Why did they feel the need to concentrate on the translator instead of the translation process? Where did they take the area of translation studies (TS) and what were their points of contention? According to Kobus Marais on the untitled book "The Routledge Handbook of the History of Translation Studies" the efforts to make translation more human centered have helped us understand how translation works in real life. This shift has connected translation studies to fields like cognitive science, psychology, neuroscience, and biology, showing that translation is not just about language but also about human experience and thought. It challenges the old idea that human thinking is purely logical and rational. However, focusing too much on the human aspect of translation has its drawbacks. It can ignore the larger systems and structures that influence translation, such as social, cultural, and institutional factors. Thinkers like Bourdieu, Giddens, and Latour have emphasized the need to balance individual choices with the broader structures that shape them. If translation studies focus only on

the translator as an individual, they risk overlooking these important systems. Another issue is the idea of the activist translator at the heart of this human-centered approach. Is this idea truly universal, or is it based on a Western, liberal way of seeing individuals? In many cultures, people may not view individual agency in the same way. Even in Western thought, newer theories about how the mind works suggest that thinking and decision-making are more distributed and collective than we once believed. Instead of focusing on just one perspective, a broader, more flexible approach that considers multiple viewpoints could be more useful in the long run. Thus, humanizing translation has opened the door to an even bigger question: what about non-human translation? Animals also translate information in their own ways, and communication happens not just between humans, but also between humans and non-humans (such as animals or even AI). This is an area that translation studies have yet to fully explore.

Today, in the era of AI the role of human translators extends far beyond simply replacing words from one language with another. Although AI translation systems have become increasingly efficient at translating large volumes of text, professional human translators remain essential in situations where accuracy, cultural understanding, and ethical judgment are required. In high-stakes fields such as law, medicine, and government, even small translation errors can have serious legal, financial, or health consequences. For this reason, human translators are responsible for ensuring that translations are accurate, comply with professional standards, and are suitable for official use.

Human translators also play an important role in preserving the cultural and emotional aspects of language. They are able to recognize idiomatic expressions, humor, irony, tone, and cultural references that AI systems may translate too literally or fail to interpret correctly. (Wei, 2024) This ability is particularly valuable in creative writing, marketing, branding, and diplomatic communication, where the intended message depends not only on linguistic accuracy but also on cultural appropriateness and audience expectations.

Rather than replacing human translators, AI has changed the way they work. Many professional translation workflows now follow a *human-in-the-loop* approach, in which AI produces an initial translation and human translators review, edit, and refine the output. During this post-editing process, translators correct terminology, improve fluency, ensure consistency, and identify potential cultural bias or factual errors.(Wang, 2023) As a result, the translator's role has evolved from producing every translation from scratch to supervising AI systems, managing translation quality, and providing linguistic and cultural expertise. This development shows that while AI has significantly improved translation speed and productivity, human translators continue to play a vital role in delivering accurate, culturally appropriate, and trustworthy translations, especially in sensitive and high-value contexts.

1.8 The significance of cultural context:

When translating a speech, context plays a crucial role in ensuring accuracy and clarity. Words cannot be translated in isolation, as both the foreign language and the native language carry similar ambiguities. Therefore, integrating context into the translation process is essential. To accurately interpret and respond to oral and written statements, one must consider the specific situation in which they are made, as every meaningful statement occurs within a particular context and serves a specific purpose. (Saeed, 2014, p. 7) Highlighting the significance of context in translation is vital, as it enhances understanding and contributes to improved accuracy and efficiency.

Recognizing the different types of contexts necessary for translation helps achieve more precise and meaningful results. If we take the sentence “I appreciate that,” it is difficult to fully understand its meaning without knowing who "I" refers to or what "that" signifies. The interpretation depends on external information, such as the speaker’s identity and the subject of appreciation. Similarly, in machine translation, relying solely on statistical memory may lead to accurate translations only if the contextual meaning is correctly understood. Additionally, some words or phrases have multiple possible meanings, a phenomenon known as ambiguity. Context plays a

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crucial role in resolving this uncertainty. Context is essential in translation and communication for several reasons:

1. It provides the necessary background to interpret a message accurately, guiding the recipient on how to understand its significance and implications.
2. Effective communication requires awareness of appropriate conversational styles in different social and cultural settings.
3. Without context, the true intent of a message can be misunderstood, leading to ineffective communication.
4. Understanding context allows individuals to navigate both verbal and non-verbal exchanges more effectively. (Saeed, 2014, p. 7)

In summary, context is a fundamental element of translation and communication, ensuring clarity and reducing misunderstandings.

Conclusion

This chapter has provided a foundational overview of the primary translation methods, contrasting the rapid technological evolution of AI translation with the enduring significance of human translation. Through a detailed examination of the historical trajectory, technological underpinnings, and operational mechanisms of AI-based system including rule-based, statistical, and neural machine translation models it becomes evident that AI has significantly advanced in its ability to replicate certain aspects of human linguistic behavior. Yet, despite these advances, numerous challenges persist, especially in the accurate handling of ambiguity, idiomatic expressions, and cultural nuance.

In parallel, this chapter has reaffirmed the central role of human translators, emphasizing their unique capacity for contextual reasoning, cultural sensitivity, and ethical judgment elements that remain beyond the reach of even the most sophisticated AI systems. Furthermore, traditional human translation methods, such as semantic, communicative, and faithful translation, continue to offer valuable insights into the practice of meaning transfer across languages.

By highlighting the interplay between these two approaches, this chapter sets the stage for the comparative analysis that will follow. The next chapter will delve into empirical data collected from Master's translation students, providing a case-based evaluation of how AI and human translation methods are perceived and applied in an academic setting. This exploration aims to inform the broader discourse on the future of translation, questioning whether AI will complement, compete with, or ultimately transform the role of human translators.

CHAPTER 02
RESEARCH
METHODOLOGY

02

Chapter Two: Research Methodology

Introduction

This chapter outlines the methodological framework employed to investigate the comparative effectiveness of artificial intelligence (AI) translation and human translation. Given the rapid integration of AI tools into the field of translation, it is essential to adopt a systematic and structured approach that evaluates their practical performance and perception among translation students. This chapter details the research design, data collection procedures, participant demographics, instruments used, and the methods of data analysis. It also highlights ethical considerations and potential limitations of the study. The aim is to provide a clear and replicable process that supports the reliability and validity of the research findings.

2.1 Research Design

In order to obtain a thorough grasp of the advantages, disadvantages, and perceived usefulness of AI versus human translation, this study uses a mixed-methods research methodology, integrating quantitative and qualitative techniques. Two main factors led to the decision that this strategy was appropriate:

The quantitative component was used to collect structured data through a questionnaire, allowing for statistical analysis of variables such as frequency of AI tool usage, accuracy ratings, satisfaction levels, and preferences. This aspect of the research helps in identifying general trends and drawing comparisons between the two translation methods.

The qualitative component focused on open-ended responses that explored the reasons behind participants' preferences, perceived strengths and weaknesses of each translation method, and expectations for future developments. This enriched the study by capturing the complexity and subjectivity inherent in language perception and translation quality.

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By combining both methods, this research design ensures a balanced and nuanced exploration of the topic. The quantitative data offers objectivity and comparability, while the qualitative data adds depth and contextual understanding. This approach aligns with the study's aim to not only measure but also interpret the evolving relationship between human expertise and artificial intelligence in translation practices.

2.2 Research Setting and Participants

This study was conducted within the Translation Department at the University of Oran 2, an academic institution known for its focus on language education, translation studies, and applied linguistics. The setting was selected for its relevance to the research topic and the accessibility it offered to a target population actively engaged in the theoretical and practical dimensions of translation. Given that this institution trains future translators, it provided a suitable environment to examine current attitudes toward AI translation tools in comparison to traditional human methods.

The participants of the study consisted of 32 individuals, primarily Master's level students, enrolled in programs related to Linguistics and Applied Languages and Literature and Civilization. The age group of respondents ranged predominantly from 20 to 27 years, with a significant portion identifying as female, reflecting the gender distribution typical of the humanities and translation studies fields.

Participants were selected using a non-probability convenience sampling method, based on their availability and willingness to complete the questionnaire. This method was deemed appropriate given the exploratory nature of the research and the focus on gathering in-depth perspectives from individuals with academic exposure to translation theory and practice.

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In terms of linguistic proficiency, most participants self-reported intermediate to advanced levels (B1–C2) in foreign languages, which positioned them well to evaluate the nuances of both AI-generated and human-produced translations. Although only a few had professional experience in translation, the majority had substantial academic training in translation tasks, equipping them with the necessary analytical skills to assess translation quality critically.

The diversity in academic focus, language proficiency, and user experience with translation technologies among participants enriched the data, allowing for a multifaceted understanding of how AI and human translation are perceived in an academic context. This participant profile not only reflects the emerging generation of translation professionals but also offers insight into how future translators might integrate AI tools into their practice or resist them.

2.3 Data Collection Tools

For the purpose of this research, data was collected using a structured online questionnaire, which served as the primary and only research instrument. The questionnaire was carefully designed to gather both quantitative and qualitative data relevant to the study's objective: to assess and compare the effectiveness, reliability, and perception of AI translation in contrast to human translation.

2.3.1 Purpose and Rationale

The use of a questionnaire was chosen due to its efficiency in reaching a relatively large number of participants within a short timeframe, as well as its capacity to collect diverse data types in a consistent format. It allowed respondents the flexibility to complete the survey at their convenience while maintaining the anonymity required to ensure honest and uninfluenced responses. The questionnaire was administered via an

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online platform (e.g., Google Forms), which facilitated digital distribution and streamlined the process of data collection and analysis.

2.3.2 Structure and Content

The questionnaire was composed of both closed-ended and open-ended questions. It was divided into several sections to address different aspects of the participants' backgrounds and their attitudes and experiences regarding AI and human translation. These sections included:

a. **Demographic Information**

Participants were asked about their age range, gender, academic level, field of study, and language proficiency level. This data helped contextualize their perspectives based on their academic and linguistic backgrounds.

b. **Professional and Academic Experience**

Respondents were asked whether they had any professional experience as translators or had received formal training in translation. This helped determine their level of familiarity with translation processes and quality evaluation.

c. **Experience with Translation Tools**

This section explored whether participants had previously used AI-powered translation tools (e.g., Google Translate, DeepL, ChatGPT), how frequently they used these tools, and whether they had any experience with human translation services, such as professional translators or agencies.

d. **Preference and Perception**

Participants were asked to indicate whether they preferred AI or human translation, and to justify their preference. This section also included a question on how well each method handled contextual meaning and idiomatic expressions an area often seen as a key differentiator between AI and human translation.

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e. Evaluation Metrics

Respondents rated the accuracy of AI translation compared to human translation on a scale from 1 to 10 and indicated their level of satisfaction with AI-generated translations. They were also asked whether speed influenced their preference and whether they would be willing to pay more for human translation.

f. Outlook and Suggestions

The final part of the questionnaire invited participants to offer suggestions for improving AI translation tools and to share their views on the future of translation services, including which method AI or human they believed would dominate the field in the future.

2.3.3 Nature of Data Collected

The questionnaire generated both numerical data (e.g., ratings and frequency measures) and descriptive data (e.g., written opinions and justifications). This dual nature allowed the research to conduct basic statistical analysis for general trends while also performing thematic analysis on open-ended responses to identify key recurring themes, such as accuracy, emotional resonance, ethical trust, or the human ability to understand context.

2.3.4 Validity and Reliability

To ensure content validity, the questions were constructed to directly align with the research questions and objectives. Each item was designed to measure a specific aspect of the participants' experiences or opinions related to translation. The clarity of language, logical flow, and relevance of each question were reviewed and revised through multiple drafts before final distribution.

Although no formal pilot test was conducted, the questionnaire was carefully structured to minimize ambiguity and guide participants smoothly from factual

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questions to more reflective and evaluative ones. The use of a digital platform also ensured standardized data recording and reduced the risk of human error in data entry.

2.4 Data Analysis

Once the questionnaires were collected, the process of understanding what they revealed began. Because this study used a mixed-methods approach, the analysis was carried out on two levels: one focusing on the measurable, numerical patterns (quantitative), and the other diving into the open-ended, reflective responses (qualitative). This dual lens allowed for a richer and more balanced interpretation of how students view and experience both AI and human translation.

1. Quantitative Insights

The structured questions the ones with clear, predefined answers provided a wealth of straightforward data. These included how often students used AI translation tools, how satisfied they were with the results (on a scale from 1 to 5), and how they rated AI's accuracy compared to human translation (on a scale from 1 to 10). Other questions looked at whether they had used professional human translation services and whether they would be willing to pay more for human translation over AI.

To make sense of these responses, standard descriptive statistical methods were used. Frequencies helped reveal how common certain responses were such as how many participants preferred AI or how many had formal translation training. Percentages showed the proportions within the group, offering a clearer picture of trends. And averages were calculated to understand general satisfaction or accuracy ratings. Together, these numbers painted a clear outline of how AI is being used and judged by students today.

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2. Qualitative Exploration

But numbers alone do not tell the full story. That is where the open-ended responses came in short texts where participants could speak freely about their preferences, opinions, and expectations. These parts were especially valuable for understanding *why* someone preferred human over AI translation, how they thought each method dealt with context and idioms, and what improvements they wished to see in AI tools.

To analyze this more subjective data, a method called thematic content analysis was used. It began with getting familiar with the responses reading them several times to absorb what was being said and begin noticing patterns. From there, key phrases and ideas were highlighted and grouped into early categories: things like "accuracy," "speed," "context," or "emotion."

As more responses were examined, these smaller ideas naturally connected into larger themes. Many participants, for example, talked about how reliable they found AI, some trusting it for quick tasks, others warning against relying on it too heavily. A recurring contrast emerged between the speed of AI and the depth of human translation, with emotional tone and cultural awareness being qualities many felt only humans could provide. There were also ethical concerns and reflections on how AI might shape or reshape the profession.

By looking at these themes as a whole, a clear narrative began to emerge. One respondent, for example, wrote:

"I prefer human translation because it's more accurate and understands meaning better than just words."

Comments like this helped give voice to the numbers. They revealed not just trends, but the reasoning and emotion behind those trends. These insights helped confirm that while AI may be useful, many students still see human translation as essential especially for tasks that require nuance, trust, or deep cultural context.

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3. Bringing the Results Together

After examining both the numbers and the words the statistics and the reflections it became clear that neither side alone could fully capture what the participants felt and believed. It was in bringing these two forms of data together that the research truly came into focus.

On the one hand, the quantitative analysis showed that many students appreciated AI translation for what it does best: speed and accessibility. It's fast, available, and gets the job done when time is limited or the task is simple. But the open-ended responses told a more layered story one that highlighted what was missing. Participants often pointed out how AI translation struggles with nuance, especially when dealing with idioms, emotional tone, or culturally sensitive content. These are the places where machines still fall short, and where the human touch remains irreplaceable.

Although AI tools were clearly part of many students' routines, they didn't replace their trust in human translation. In fact, the more serious or formal the context like academic writing or emotionally meaningful content the more students leaned toward human expertise. It wasn't just about getting the words right; it was about understanding what those words meant within their context.

Putting all this together created a fuller, more balanced picture of where AI stands in the world of translation. It highlighted not only what AI is capable of today, but also what it still lacks. Most importantly, it laid the foundation for deeper discussions on what the future might look like how technology and human skill might continue to develop side by side.

2.5 Ethical Considerations

Ensuring the ethical integrity of the research process was a priority in the design and implementation of this study. As the research involved human participants specifically students providing personal opinions and experiences it was essential to uphold ethical

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standards that protect participant rights, promote transparency, and ensure the responsible use of data.

2.5.1 Informed Consent

Participation in the questionnaire was entirely voluntary. All participants were clearly informed, at the beginning of the questionnaire, about the purpose of the study, the nature of their participation, and their rights as research subjects. They were explicitly notified that:

- Their responses would be used solely for academic research.
- They could choose to skip any question they were uncomfortable answering.
- They could withdraw from the questionnaire at any point before submission without consequence.

By proceeding with the questionnaire, participants indicated their informed consent to take part in the study.

2.5.2 Anonymity and Confidentiality

No personally identifiable information (such as names, student IDs, or contact details) was collected in the questionnaire. This measure ensured anonymity, allowing participants to respond freely without fear of judgment or repercussions.

The data collected was stored securely in password-protected digital files accessible only to the researcher. All responses were analyzed and presented in aggregate form, and any direct quotes included in the study were anonymized to protect the identity of respondents.

2.5.3 Data Integrity and Use

The data was collected, processed, and analyzed with the utmost care to ensure accuracy and integrity. There was no manipulation of responses, and every effort was

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made to represent participants' views faithfully. The findings were interpreted objectively and used solely to support the academic goals of this dissertation.

No part of the data has been or will be shared with third parties outside the academic context of this research.

2.5.4 Researcher's Responsibility

As the sole researcher, I have taken full responsibility for maintaining ethical standards throughout the research process. This includes designing neutral and non-leading questions, avoiding coercion, and interpreting results with fairness and respect for differing viewpoints. The study was conducted in accordance with the general ethical guidelines of academic research and the norms observed at the University of Oran 2.

Conclusion

This chapter has outlined the methodological framework adopted to explore the comparative effectiveness and perception of AI and human translation methods among translation students at the University of Oran 2. It detailed the rationale behind the mixed-methods approach, the structure and purpose of the questionnaire, and the procedures for data analysis, while also acknowledging the ethical considerations and inherent limitations of the study. The insights derived from this methodology form the empirical foundation for the following chapter, which presents and analyzes the key findings of the research in relation to the initial research questions and objectives.

CHAPTER 03
FINDINGS
AND
DISCUSSION

03

Chapter Three: Findings and Discussion

3.1 Introduction

This chapter presents and discusses the empirical findings generated by the questionnaire administered to a sample of thirty-two ($N = 32$) students affiliated with the Department of English at the University of Oran 2. The instrument was designed to elicit both quantifiable data regarding respondents' demographic and academic profiles, their patterns of engagement with artificial intelligence (AI) translation tools, and their evaluative judgments of machine-generated output relative to human translation, as well as open-ended qualitative responses capturing the reasoning that underlies these judgments. The overarching objective of this chapter is to organise, describe, and interpret this body of data in a manner that directly addresses the central research question guiding the present study, namely whether artificial intelligence, in its current state of development, is capable of replacing human expertise in the field of translation.

The chapter is structured according to a separated analytical approach that distinguishes between the numerical and the discursive dimensions of the data before integrating them into a unified interpretive discussion. Section 3.2 presents the quantitative findings, reporting the descriptive statistics obtained from the closed-ended items of the questionnaire in tabular form, each table being accompanied by an objective description and interpretation of the reported percentages. Section 3.3 turns to the qualitative findings, organising the open-ended responses into three recurrent thematic categories and illustrating each theme through direct participant quotations. Section 3.4 constitutes the discussion proper, in which the quantitative and qualitative datasets are read against one another in order to formulate a coherent answer to the thesis question. Section 3.5 acknowledges the methodological limitations that condition the scope and generalisability of the findings, and Section 3.6 offers a concluding synthesis of the chapter together with a transitional statement leading into the General Conclusion.

3.2 Quantitative Findings

The quantitative component of the questionnaire yielded descriptive data across three broad domains: the demographic and linguistic stratum of the respondent pool, the

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patterns of workflow integration and adoption of AI translation tools, and the evaluative quality metrics through which respondents rated their perceptions of machine translation relative to human translation. Each domain is presented below in the form of frequency and percentage tables, followed by an interpretive commentary.

3.2.1 Demographic and Linguistic Profile of Respondents

Table 3.1: Age Distribution of Respondents (N = 32)

Age Group	Frequency	Percentage
20–27 years	31	96.88%
Under 20 years	1	3.12%
Total	32	100.00%

As indicated in Table 3.1, the overwhelming majority of respondents, thirty-one out of thirty-two participants (96.88%), fall within the 20–27 age bracket, while a single respondent (3.12%) is under the age of twenty. This near-total concentration within a single age cohort situates the sample firmly within the demographic of young adults who have completed, or are in the process of completing, higher education, and who may be presumed to have reached adulthood during a period in which digital and AI-based tools were already integrated into academic and professional life. The homogeneity of the age distribution suggests that the attitudes captured throughout the remainder of the questionnaire are largely representative of a single generational perspective rather than a cross-generational comparison, a point that bears directly on the interpretation of the technology-adoption patterns discussed in subsequent sections.

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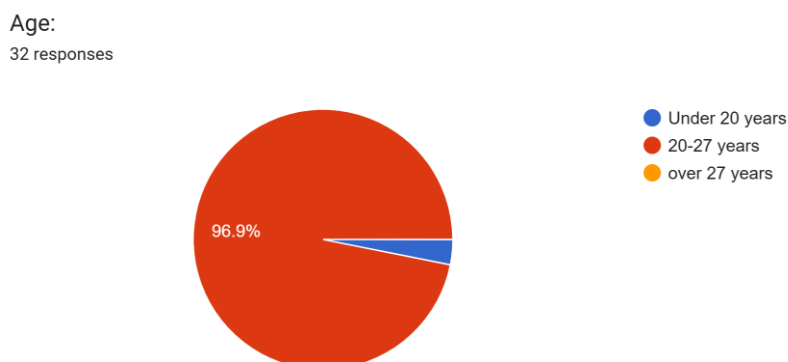


Figure 3.1: Age Distribution of Respondents (N = 32)

Table 3.2: Gender Distribution of Respondents (N = 32)

Gender	Frequency	Percentage
Female	28	87.50%
Male	4	12.50%
Total	32	100.00%

Table 3.2 reveals a pronounced gender asymmetry within the sample, with female respondents constituting the substantial majority (87.50%, n = 28) and male respondents representing a considerably smaller proportion (12.50%, n = 4). This imbalance is broadly consistent with the gender composition typically observed in language and translation programmes within Algerian universities, where female enrolment tends to predominate. While this distribution was not a deliberate sampling criterion, it should nonetheless be borne in mind when generalising the findings, since the perceptions reported in this chapter are disproportionately shaped by the views of female students of translation and applied languages.

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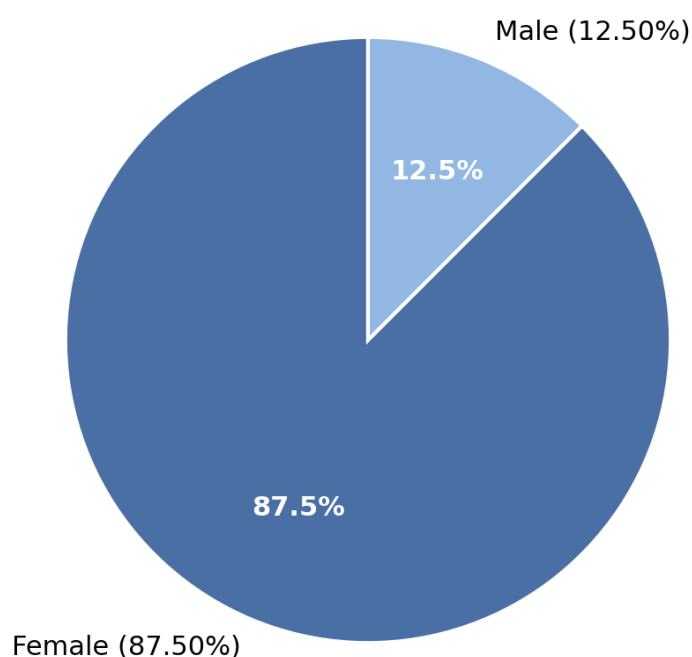


Figure 3.2: Gender Distribution of Respondents (N = 32)

Table 3.3: Educational Level of Respondents (N = 32)

Educational Level	Frequency	Percentage
Master	28	87.50%
Licence	3	9.38%
PhD	1	3.12%
Total	32	100.00%

With regard to educational attainment, Table 3.3 shows that the sample is dominated by Master's-level students, who account for 87.50% ($n = 28$) of respondents, while Licence-level students represent 9.38% ($n = 3$) and a single PhD candidate contributes the remaining 3.12%. This concentration at the postgraduate level indicates that the respondents, for the most part, possess an intermediate to advanced level of academic training in language studies, and are therefore likely to have encountered translation as

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both an academic subject and a practical skill prior to their participation in the study. The near-absence of undergraduate voices and the marginal presence of doctoral-level input mean that the findings should be read primarily as reflecting the perspective of Master's students, a population that occupies a transitional position between formal instruction and professional practice.

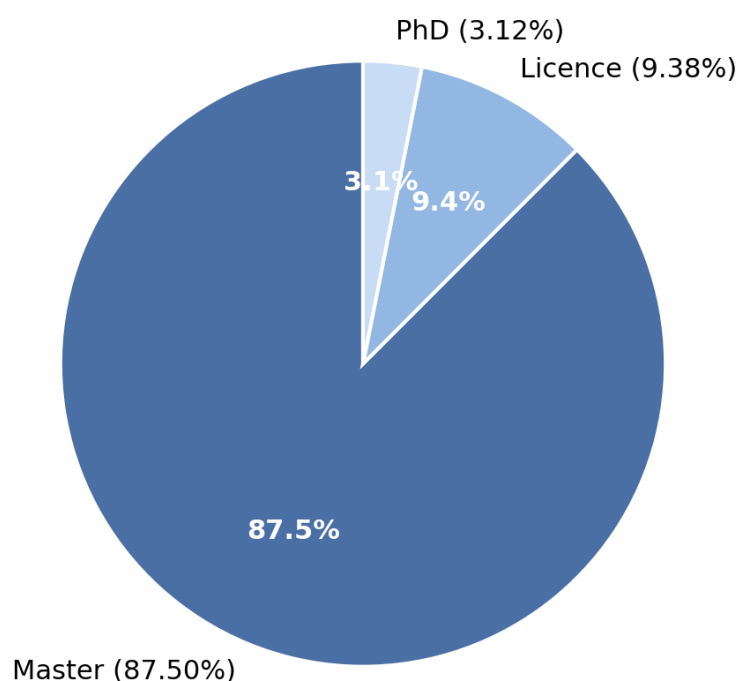


Figure 3.3: Educational Level of Respondents (N = 32)

Table 3.4: Field of Study of Respondents (N = 32)

Field of Study	Frequency	Percentage
Linguistics and Applied Languages	24	75.00%
Literature and Civilization	3	9.38%
Other Fields	5	15.62%
Total	32	100.00%

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Table 3.4 indicates that three-quarters of the respondents (75.00%, $n = 24$) are enrolled in Linguistics and Applied Languages, a specialisation that typically entails direct and sustained engagement with translation theory and practice. A smaller proportion of respondents report a background in Literature and Civilization (9.38%, $n = 3$), while 15.62% ($n = 5$) indicate affiliation with other, unspecified fields of study. The predominance of respondents trained in applied linguistics lends a measure of disciplinary credibility to the sample's judgments regarding translation quality, insofar as these respondents may be assumed to possess at least a foundational metalinguistic vocabulary for evaluating issues such as register, idiomaticity, and contextual accuracy, even where they have not undergone formal professional certification as translators.

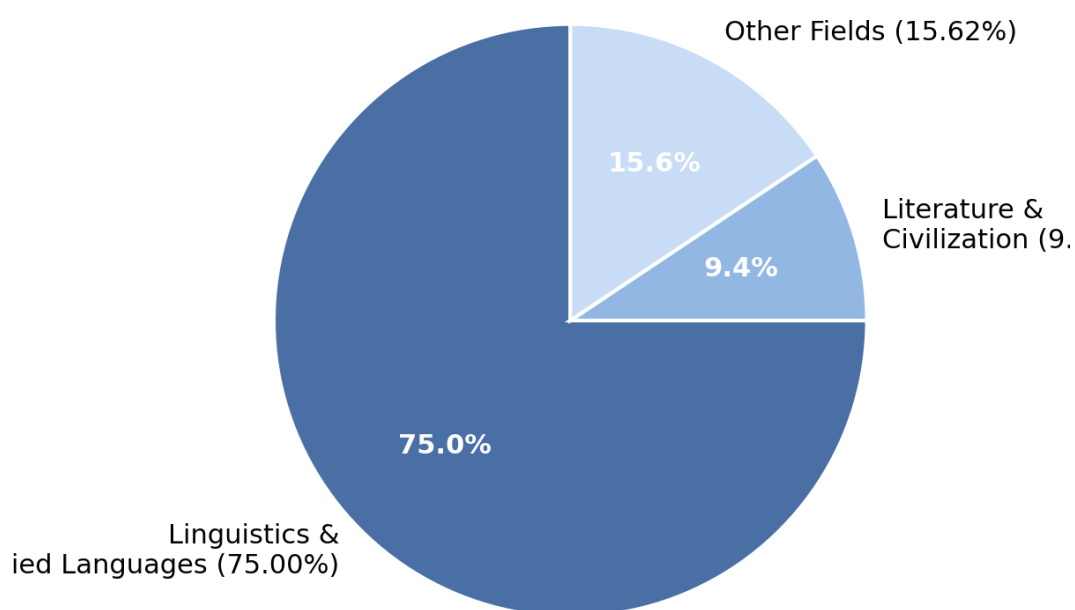


Figure 3.4: Field of Study of Respondents ($N = 32$)

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Table 3.5: Language Proficiency Level of Respondents (CEFR) (N = 32)

Proficiency Level	Frequency	Percentage
Advanced/Proficient (C1–C2)	17	53.12%
Intermediate (B1–B2)	15	46.88%
Total	32	100.00%

Finally, Table 3.5 shows a roughly balanced, though slightly skewed, distribution of self-reported language proficiency according to the Common European Framework of Reference for Languages (CEFR), with 53.12% (n = 17) of respondents situating themselves at the Advanced or Proficient level (C1–C2) and 46.88% (n = 15) at the Intermediate level (B1–B2). The near-even split between these two proficiency bands suggests that the sample is not composed exclusively of highly advanced language users, and that judgments concerning the adequacy of AI-generated translations are formed by individuals with a range of linguistic competence. This distribution is analytically significant, since it implies that perceptions of AI translation quality reported later in this chapter cannot be attributed solely to the discernment of an elite subgroup of near-native speakers, but rather reflect a spectrum of linguistic sensitivity extending from intermediate to advanced levels.

What's your language proficiency level?
32 responses

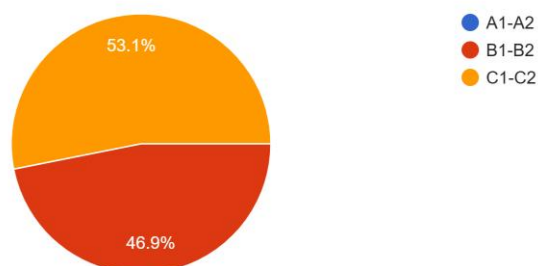


Figure 3.5: Language Proficiency Level of Respondents, CEFR (N = 32)

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3.2.2 Workflow Integration and Adoption Patterns

Table 3.6: Formal Translation Training or Professional Experience (N = 32)

Response	Frequency	Percentage
No / Never	26	81.25%
Yes / Conditional	6	18.75%
Total	32	100.00%

Table 3.6 demonstrates that the substantial majority of respondents, 81.25% (n = 26), report having received no formal translation training and no prior professional experience in the field, while only 18.75% (n = 6) indicate some form of training or conditional professional exposure. This finding is of considerable importance for the interpretation of subsequent data, as it establishes that the sample's attitudes toward AI and human translation are formed predominantly by individuals situated outside the professional translation industry. Their evaluations, in other words, are grounded less in codified professional standards of translation quality than in the practical, lived experience of using translation as language learners and academic consumers of bilingual or multilingual content, a positioning that arguably renders them representative of the broader population of everyday AI translation users rather than of specialist translators.

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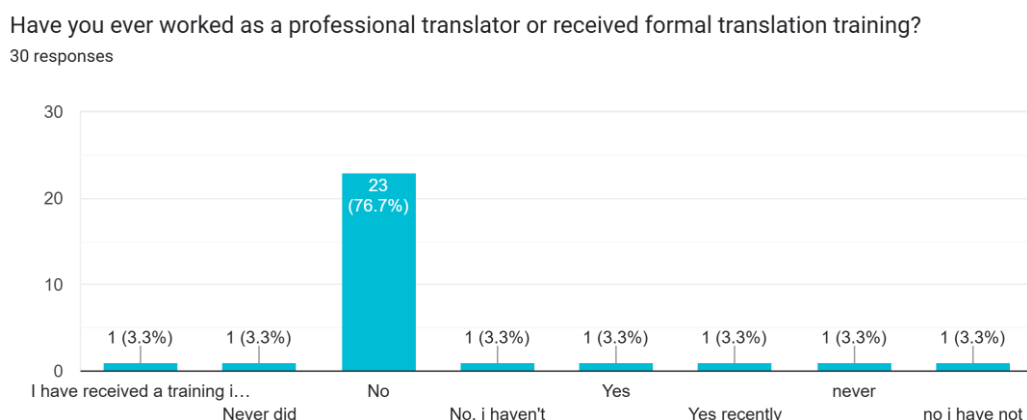


Figure 3.6: Formal Translation Training or Professional Experience (N = 32)

Table 3.7: Prior Adoption of AI Translation Tools (N = 32)

Response	Frequency	Percentage
Yes	31	96.88%
Not Very Often	1	3.12%
Total	32	100.00%

As shown in Table 3.7, prior adoption of AI translation tools is virtually universal within the sample, with 96.88% (n = 31) of respondents affirming that they have used such tools, and only a single respondent (3.12%) reporting infrequent use. This near-total penetration confirms that AI-assisted translation has become a normalised component of the academic and linguistic routines of the respondent population, rather than a marginal or experimental practice. The near-ceiling adoption rate also indicates that the evaluative and attitudinal data reported elsewhere in this chapter are grounded in first-hand, experiential familiarity with AI translation tools rather than in speculative or hypothetical impressions, thereby strengthening the empirical weight of the perceptions subsequently analysed.

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Have you ever used AI-powered translation tools (e.g., Google Translate, DeepL, ChatGPT, etc.)?
32 responses

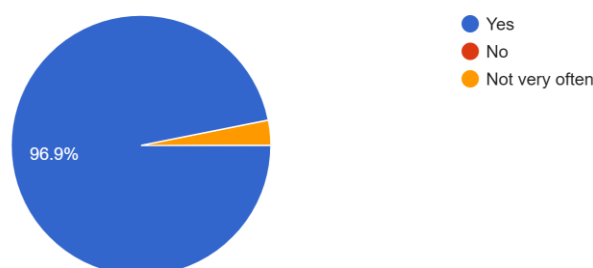


Figure 3.7: Prior Adoption of AI Translation Tools ($N = 32$)

Table 3.8: Frequency of AI Translation Tool Use ($N = 32$)

Frequency	Frequency (n)	Percentage
Sometimes	26	81.25%
Always	5	15.62%
Missing Response	1	3.12%
Total	32	100.00%

Table 3.8 refines the adoption pattern reported above by detailing the frequency with which AI translation tools are used. The majority of respondents, 81.25% ($n = 26$), describe their use as occasional, selecting the option "sometimes", while a smaller proportion, 15.62% ($n = 5$), report using such tools consistently, selecting "always"; one response (3.12%) was not recorded. This distribution suggests that although AI translation tools have achieved near-universal adoption, as established in Table 3.7, their integration into respondents' workflows remains predominantly supplementary rather than exclusive. The comparatively low proportion of habitual, constant users implies that respondents tend to treat AI tools as an auxiliary resource to be consulted selectively, rather than as a wholesale substitute for their own linguistic judgment, a

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pattern that foreshadows the more ambivalent evaluative findings discussed in Section 3.2.3.

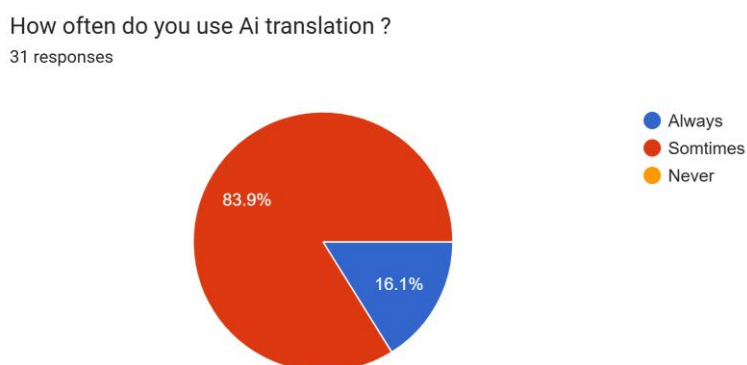


Figure 3.8: Frequency of AI Translation Tool Use ($N = 32$)

Table 3.9: Prior Interaction with Paid or Human Translation Services ($N = 32$)

Response	Frequency	Percentage
No	21	65.62%
Yes	10	31.25%
Missing Response	1	3.12%
Total	32	100.00%

Table 3.9 indicates that a majority of respondents, 65.62% ($n = 21$), have never engaged the services of a paid human translator, whereas 31.25% ($n = 10$) report having done so, and one response (3.12%) is missing. This finding suggests that direct comparative experience with professional human translation services remains relatively limited within the sample, a circumstance that carries implications for the interpretation of respondents' comparative judgments between AI and human translation. Where such judgments are not grounded in direct professional experience, they are more likely to be informed by classroom instruction, informal exposure to bilingual texts, or generalised

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assumptions about the capabilities of human translators, rather than by first-hand evaluation of commissioned, professionally rendered translations.

Have you ever used human translation services (e.g., professional translators, freelancers, or agencies)?
31 responses

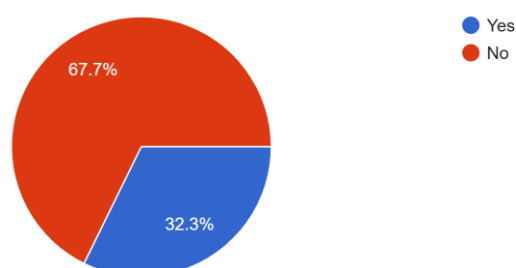


Figure 3.9: Prior Interaction with Paid or Human Translation Services (N = 32)

3.2.3 Evaluative Quality Metrics

Table 3.10: Perceived AI Accuracy Relative to Human Translation (Scale of 1–10)

Score	Frequency	Percentage
4	2	6.25%
5	4	12.50%
6	11	34.38%
7	5	15.62%
8	4	12.50%
9	5	15.62%
10	1	3.12%

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Score	Frequency	Percentage
Total	32	100.00%

Table 3.10 reports respondents' ratings of the perceived accuracy of AI translation relative to human translation on a scale from one to ten. No respondent selected a score below four, indicating that outright rejection of AI translation accuracy is absent from the sample; however, the distribution is markedly concentrated at the mid-range, with the modal score of 6 accounting for 34.38% ($n = 11$) of responses, the single largest proportion recorded on the scale. Scores of 7, 8, and 9 collectively account for 43.76% ($n = 14$) of responses, indicating a substantial secondary cluster of respondents who regard AI accuracy relatively favourably, while scores of 4 and 5 together account for 18.75% ($n = 6$) of more sceptical evaluations. Only one respondent (3.12%) awarded the maximum score of 10. The overall configuration of this distribution, clustering heavily around a moderate score of 6 rather than dispersing evenly or gravitating toward the upper end of the scale, indicates that respondents perceive AI translation as functionally competent but not yet equivalent in accuracy to human translation, a nuanced middle-ground judgment that sets the stage for the discussion in Section 3.4.

How would you rate the accuracy of AI translation compared to human translation on a scale of 1-10?

32 responses

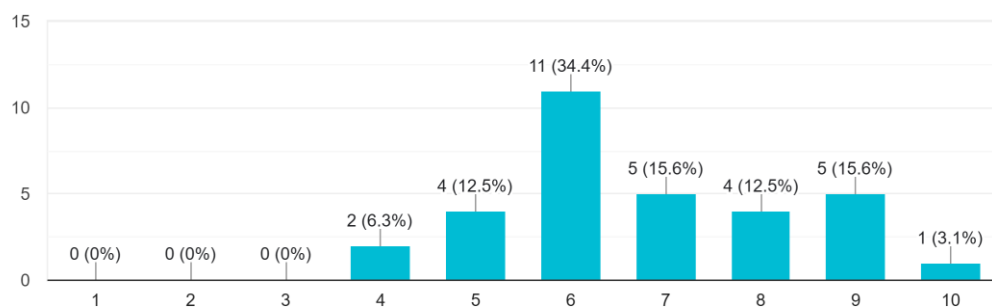


Figure 3.10: Perceived AI Accuracy Relative to Human Translation, Scale 1-10 ($N = 32$)

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Table 3.11: General Satisfaction with AI Translation Relative to Human Translation (Scale of 1–5) (N = 32)

Score	Descriptor	Frequency	Percentage
1	Not Satisfied	1	3.12%
2	—	3	9.38%
3	Moderately Satisfied / Neutral	15	46.88%
4	Satisfied	12	37.50%
5	Very Satisfied	1	3.12%
Total	—	32	100.00%

Table 3.11 presents respondents' general satisfaction with AI translation relative to human translation on a five-point scale. The most frequently selected response is the neutral midpoint, "moderately satisfied", chosen by 46.88% (n = 15) of respondents, followed by "satisfied" at 37.50% (n = 12). Together, these two categories account for 84.38% of the sample, indicating that the substantial majority of respondents locate their satisfaction with AI translation somewhere between cautious neutrality and moderate approval, without extending into unreserved endorsement. The extremes of the scale are sparsely populated, with only 3.12% (n = 1) reporting dissatisfaction and an equal proportion (3.12%, n = 1) reporting very high satisfaction, while a further 9.38% (n = 3) selected the second point on the scale. This concentration at the neutral-to-moderately-satisfied range, mirroring the mid-range clustering already observed in Table 3.10, constitutes one of the most analytically significant findings of the quantitative dataset, and is addressed in detail in the discussion that follows.

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How satisfied are you with the results of AI translation versus human translation?

32 responses

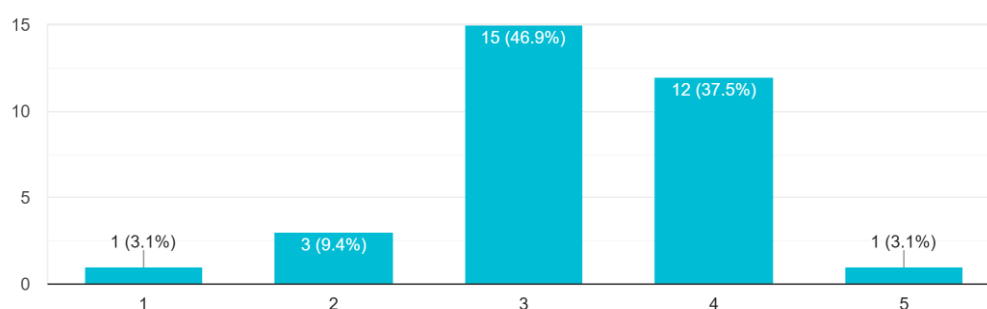


Figure 3.11: General Satisfaction with AI Translation Relative to Human Translation, Scale 1-5 (N = 32)

3.3 Qualitative Findings

The qualitative data derived from the open-ended items of the questionnaire were subjected to thematic analysis, from which three principal and recurrent themes emerged: the tension between processing speed and contextual fidelity, the tension between the cost-free efficiency of AI tools and the premium value attributed to human translation, and a cluster of systemic structural recommendations directed at the improvement of machine translation software. Each theme is discussed below and illustrated through direct quotations extracted from participant responses.

3.3.1 Theme A: Speed versus Contextual Fidelity

The most pervasive theme to emerge from the qualitative data concerns a perceived trade-off between the processing speed offered by AI translation tools and the contextual and cultural fidelity associated with human translation. Respondents who expressed a preference for AI translation grounded their justification almost exclusively in considerations of velocity and convenience, offering comparatively terse explanations that foreground efficiency over any claim to superior linguistic quality.

“AI translation because it is very fast.”

“Ai it’s easy and faster.”

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The brevity and functional orientation of these justifications is itself analytically revealing: respondents favouring AI translation rarely invoked considerations of accuracy, nuance, or stylistic quality, suggesting that AI's principal appeal for this subset of the sample lies in its capacity to reduce time and cognitive effort rather than in any conviction regarding its linguistic superiority.

By contrast, respondents who expressed a preference for human translation articulated more elaborate justifications, consistently centred on the capacity of human translators to navigate sociocultural constructs, semantic nuance, and emotional variance that AI systems are perceived to handle inconsistently.

“Human, AI has inconsistent performance, the Large Language it's properly trained when it comes to cultural and social aspects that construct any Language.”

“I prefer human translation because humans understand context, emotions, and cultural differences better than AI. While AI is fast and useful, it can sometimes make mistakes or miss the deeper meaning of a text.”

These responses articulate a more complex evaluative logic than those favouring AI, explicitly weighing AI's acknowledged advantages in speed and utility against its perceived deficiencies in the interpretive dimensions of translation, namely the rendering of emotional tone, cultural embeddedness, and contextual meaning. The co-occurrence of praise and critique within the same responses, exemplified in the second quotation above, indicates that respondents do not construct AI and human translation as categorically opposed, but rather as tools suited to different communicative demands, with speed positioned as AI's principal currency and contextual fidelity positioned as the enduring currency of human expertise.

3.3.2 Theme B: Premium Costs versus Cost-Free Efficiency

The second theme to emerge concerns respondents' attitudes toward the economic dimension of translation, specifically the question of whether the accuracy attributed to human translation justifies its financial cost relative to the largely cost-free availability

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of AI translation tools. Respondents who considered payment for human translation unnecessary grounded their position in a perceived sufficiency of AI tools for their own purposes.

“No, i can use the AI translation correctly to have the suitibal translation.”

“Not really, Ai translation helps me more in my field.”

These responses frame the question of payment in pragmatic, task-oriented terms: for respondents adopting this position, the adequacy of AI translation is assessed relative to their own competence in operating and correcting such tools, or relative to the specific demands of their field of study, rather than relative to an abstract standard of translation excellence. The implicit suggestion is that AI's cost-free efficiency renders payment for human translation a matter of discretionary preference rather than functional necessity for these particular users.

Conversely, respondents who affirmed a willingness to pay for human translation grounded their position in judgments of accuracy, naturalness, and the anticipated future scarcity value of human translation expertise in a market increasingly dominated by automated tools.

“Sure, because it's more optimal and can possibly become rare if AI take over the field.”

“Yes, because human translation is more accurate, natural, and understands context better.”

The first of these quotations is particularly noteworthy in that it introduces an explicitly forward-looking, market-oriented rationale absent from the other responses cited in this section: the respondent frames the premium value of human translation not solely in terms of present-day quality, but in terms of an anticipated future scarcity engendered by continued AI expansion within the profession. This response, together with the second quotation's emphasis on accuracy and naturalness, indicates that willingness to pay for human translation is underpinned by a dual rationale combining assessments of

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current qualitative superiority with speculative assessments of the translator's future professional positioning.

3.3.3 Theme C: Systemic Structural Recommendations for Machine Translation Software

The third theme, distinct from the comparative and economic framing of the previous two, consists of respondents' direct recommendations for the structural improvement of AI translation software. These recommendations converge overwhelmingly on a single core deficiency: the tendency of current systems toward literal, word-for-word decoding at the expense of holistic textual comprehension.

“To understand the text and translate it as a whole and not word by word.”

“AI translation tools should improve context understanding, handle idioms better, and recognize cultural nuances. They should also refine grammar and tone to make translations more natural and accurate.”

The first quotation succinctly diagnoses what respondents perceive as the fundamental architectural limitation of current AI translation systems, namely a processing logic that proceeds at the level of the individual lexical unit rather than at the level of the text as an integrated whole. The second quotation elaborates this diagnosis into a more granular set of recommendations, specifying context understanding, idiomatic competence, cultural nuance recognition, and the refinement of grammar and tone as discrete areas requiring improvement. Read together, these recommendations indicate that respondents' critique of AI translation is not merely evaluative but constructively diagnostic: rather than dismissing AI translation as categorically inadequate, respondents articulate a reasonably sophisticated, if informally expressed, understanding of the specific linguistic and pragmatic competencies that current systems have yet to master, namely those competencies most closely associated with discourse-level and sociopragmatic dimensions of meaning rather than with sentence-level or lexical accuracy.

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3.4 Discussion

The quantitative and qualitative datasets presented above, when read in conjunction, converge upon a central and analytically productive paradox that lies at the heart of this study's research question. On one hand, the quantitative findings establish that AI translation tools have achieved near-total adoption within the sample, with 96.88% of respondents reporting prior use and the overwhelming majority integrating such tools into their academic and linguistic routines with a frequency described as at least occasional. On the other hand, the same respondents cluster their evaluative judgments of AI translation at markedly moderate levels, with perceived accuracy centring on a modal score of 6 out of 10 and general satisfaction centring on a neutral-to-moderate midpoint of 3 out of 5. This configuration, whereby near-universal behavioural adoption coexists with only moderate evaluative endorsement, cannot be attributed to inconsistency or confusion on the part of respondents; rather, it reflects a coherent and empirically grounded distinction between AI's instrumental utility and its qualitative sufficiency.

This paradox is rendered fully intelligible only when the quantitative clustering is read through the interpretive lens supplied by the qualitative themes discussed in Section 3.3. The frequency data indicate that respondents use AI translation tools regularly, yet the frequency of use is itself revealing: the modal response across the sample was "sometimes" rather than "always", suggesting a pattern of selective, task-contingent consultation rather than wholesale reliance. This pattern aligns directly with Theme A, in which respondents consistently characterised AI's principal value as residing in speed and convenience rather than in interpretive depth. The evidence thus supports a functional bifurcation in respondents' mental models of translation technology: AI is engaged as an accelerator, a tool mobilised to compress the time and effort required to produce a preliminary or provisional rendering of a text, while the tasks of verifying, contextualising, and culturally calibrating that rendering are implicitly reserved for human judgment, whether the respondent's own or that of a professional translator.

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This accelerator function is further corroborated by Theme B, in which respondents' economic reasoning revealed that the perceived sufficiency of free AI tools was framed in relation to respondents' own capacity to operate and correct those tools, rather than in relation to an assumption that AI output could be trusted uncritically. The respondent who reported being able to "use the AI translation correctly to have the suitable translation" implicitly positions themselves, rather than the AI system, as the final locus of quality control, a positioning entirely consistent with the moderate rather than high accuracy and satisfaction scores reported in Tables 3.10 and 3.11. Conversely, those respondents willing to pay for human translation grounded their position in precisely the competencies that Theme C identifies as AI's principal current deficiencies, namely contextual understanding, idiomatic sensitivity, and cultural nuance, indicating that the premium value attached to human translation is not a diffuse or nostalgic preference but a targeted response to specific, articulable shortcomings.

Taken together, the two datasets support the interpretive framework of AI as an accelerator and the human translator, whether the respondent themselves or a professional third party, as an indispensable cultural and contextual filter. Within this framework, AI translation performs a genuinely valuable function, one substantial enough to warrant its near-universal adoption within the sample, but this function is understood by respondents as fundamentally complementary to, rather than substitutive of, human linguistic and cultural competence. The moderate clustering observed in the evaluative metrics of Section 3.2.3 is therefore not evidence of ambivalence or indecision, but rather the quantitative signature of a nuanced and internally consistent position: AI translation is good enough to be used constantly for provisional, low-stakes, or exploratory purposes, yet not good enough to be trusted, unsupervised, with tasks that require the accurate rendering of emotional tone, idiomatic expression, or culturally embedded meaning.

In relation to the central research question of this thesis, whether artificial intelligence can replace human expertise in translation, the combined evidence of this chapter yields a qualified and empirically grounded answer. The data do not support the proposition

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that AI has replaced, or is currently positioned to fully replace, human translation expertise; the sustained emphasis, across both the quantitative and qualitative datasets, on AI's limitations in handling context, culture, and nuance directly contradicts any claim to full substitutability. At the same time, the data equally do not support a dismissive or technologically conservative position that denies AI any legitimate role within the translation process. Instead, the evidence points toward a model of functional complementarity, in which AI translation tools are absorbed into the translation workflow as instruments of acceleration and first-pass processing, while the tasks of contextual calibration, cultural mediation, and final quality assurance remain, in the perception of this sample at least, the province of human expertise. The future of translation, as reflected in the attitudes of this cohort of emerging language professionals, therefore appears to lie not in outright substitution but in a reconfigured division of labour between human and machine.

3.5 Limitations of the Study

Several methodological limitations condition the scope and generalisability of the findings presented in this chapter and warrant explicit acknowledgment. First, the sample size ($N = 32$) is modest, and while it permits meaningful descriptive analysis, it does not support robust inferential generalisation to the broader population of translation students or AI translation users. Second, the sample is geographically and institutionally bounded to a single site, the University of Oran 2, which limits the extent to which the findings can be extrapolated to students at other Algerian universities or to translation communities operating within different linguistic, educational, or cultural contexts. Third, the demographic composition of the sample, most notably its pronounced skew toward female respondents (87.50%) and toward Master's-level students (87.50%), further narrows the population to which the findings may reasonably be generalised, and raises the possibility that the attitudes reported here are not fully representative of male students, undergraduate students, or students situated outside the fields of linguistics and applied languages.

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A further limitation concerns the nature of the data-collection instrument itself. All quantitative and qualitative data reported in this chapter are self-reported, and are therefore susceptible to the range of biases associated with self-report methodologies, including social desirability bias, imprecise self-assessment of language proficiency, and inconsistent recollection of the frequency and context of prior AI tool use. Respondents' evaluative ratings of AI accuracy and satisfaction, in particular, reflect subjective impressions rather than externally validated assessments. Finally, and most significantly, the present study did not incorporate any empirical performance testing of AI translation output against expert-rated benchmarks or gold-standard human translations. Consequently, the moderate accuracy and satisfaction scores reported in Section 3.2.3, and the thematic content analysed in Section 3.3, describe respondents' perceptions of AI translation quality rather than an objectively measured index of that quality. Future research incorporating controlled, criterion-referenced evaluation of AI-generated translations would be required to corroborate, or to qualify, the perceptual findings reported here.

3.6 Conclusion

This chapter has presented and discussed the quantitative and qualitative findings generated by a questionnaire administered to thirty-two students of the University of Oran 2, with the aim of assessing whether artificial intelligence, in its present state of development, can be considered a viable replacement for human expertise in translation. The quantitative findings established that AI translation tools are used almost universally within the sample, yet are evaluated at only moderate levels of perceived accuracy and satisfaction. The qualitative findings, organised around the themes of speed versus contextual fidelity, premium cost versus cost-free efficiency, and structural recommendations for machine translation software, provided the interpretive substance necessary to explain this apparent paradox, revealing a consistent respondent position in which AI is valued as an accelerator of the translation process while human expertise continues to be regarded as indispensable for the accurate mediation of context, culture, and nuance. The discussion in Section

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3.4 synthesised these findings into a model of functional complementarity between AI and human translation, and offered a qualified response to the thesis question, namely that AI has not replaced, and does not at present appear positioned to replace, human translation expertise, but rather is being incorporated into a reconfigured, collaborative model of translation practice. Having established these empirical findings and their interpretation, the General Conclusion that follows draws together the theoretical foundations of Chapter One, the methodological framework of Chapter Two, and the present findings, considering their combined significance for translator training, professional practice, and future research.

General Conclusion

As the translation landscape evolves, shaped by technological innovation and shifting human needs, this research set out to explore a central question: Can AI-powered translation tools replace human expertise, or do they simply complement it? The journey through this study guided by theory, method, and student insight has shown that while AI offers speed and growing sophistication, it still falls short of replicating the depth, sensitivity, and contextual awareness that define human translation.

The theoretical foundations helped distinguish the mechanical from the meaningful. Where AI relies on algorithms and data patterns, human translation remains grounded in intent, interpretation, and culture. The gap between the two is not just technical it is conceptual. Human translators bring emotional nuance, ethical judgment, and cultural literacy that AI has yet to convincingly emulate.

The research methodology, centered on a case study at the University of Oran 2, allowed for a layered exploration through a mixed-methods approach. Participants, primarily Master's students in translation studies, offered both structured responses and personal reflections on their experiences with AI and human translation. Their feedback, analyzed both quantitatively and qualitatively, revealed recurring themes: AI is fast and practical, but often superficial. Human translation, while slower, is more trustworthy especially in sensitive, creative, or complex contexts.

The findings make one point especially clear: AI is not a replacement but a resource. It is a tool with increasing potential, yet it is still dependent on human oversight, particularly where precision and nuance are critical. Participants imagined a future not where AI takes over, but where it assists where machines handle routine tasks and humans bring meaning, emotion, and insight to the final product.

This research doesn't present a simple answer, but rather a balanced understanding: the future of translation lies not in choosing between AI and human translators, but in learning how to integrate both. Together, they can create a more efficient, accessible,

and deeply human form of communication where technology enhances, rather than erases, the role of the translator.

APPENDICES

Appendix “A”: The students’ Questionnaire

The Future of Translation: Can AI Replace Human Expertise?

This questionnaire aims to explore the differences between AI-powered and human translation in terms of accuracy, efficiency, and user satisfaction. As technology continues to evolve, AI translation tools are becoming increasingly common, raising questions about their reliability compared to traditional human translators. Through this survey, we seek to gather insights on user experiences, preferences, and perceptions regarding both methods. Your responses will help analyze the strengths and limitations of each approach and contribute to a better understanding of the future of translation. Thank you for your participation!

1. **Age:**

- ☐ Under 20 years
- ☐ 20-27 years
- ☐ over 27 years

2. **Gender:**

- ☐ Female
- ☐ Male

3. **Educational level?**

- ☐ Bachelor
- ☐ Masters
- ☐ Phd

4. **What's your field of study?**

- ☐ Translation
- ☐ Linguistics AAL
- ☐ Literature
- ☐ **Others**

5. **What's your profession ?**



6. **What's your language proficiency level?**

- ☐ A1-A2
- ☐ B1-B2

- C1-C2

7. **Have you ever worked as a professional translator or received formal translation training?**



8. **Have you ever used AI-powered translation tools (e.g., Google Translate, DeepL, ChatGPT, etc.)?**

- Yes
- No

9. **How often do you use Ai translation ?**

- Always
- Sometimes
- Never

10. **Have you ever used human translation services (e.g., professional translators, freelancers, or agencies)?**

- Yes
- No

11. **Do you prefer AI translation or human translation? Why?**



12. **How would you rate the accuracy of AI translation compared to human translation on a scale of 1-10?**

- 1-10

13. **Which method do you find faster, and does speed influence your preference?**



14. **How well do you think each method handles context and idiomatic expressions?**



15. **How satisfied are you with the results of AI translation versus human translation?**

- ★ *****

16. **Are you willing to pay more for human translation, and why?**



17. What improvements would you suggest for AI translation tools?



18. How do you see the future of translation services and which method do you think will dominate?



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